Pumping Lemma for Context-free Languages

Original Slides by C. Busch, LSU. Modified by A. H. Özer, Marmara University.

Take an **infinite** context-free language



Generates an infinite number of different strings

Example:

$$S \to ABE \mid bBd$$

$$A \to Aa \mid a$$

$$B \to bSD \mid cc$$

$$D \to Dd \mid d$$

$$E \to eE \mid e$$

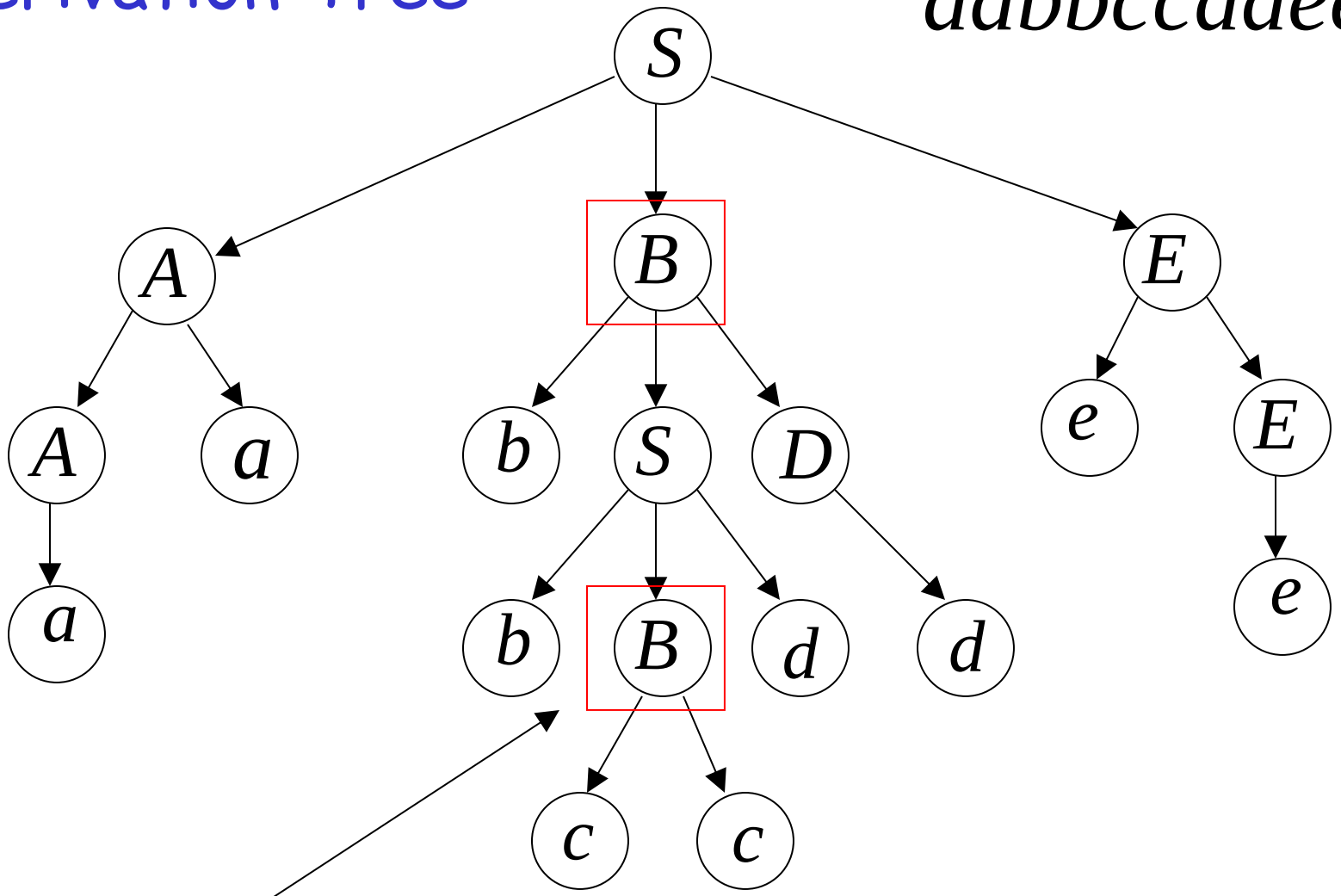
In a derivation of a "long" enough string, variables are repeated

A possible derivation:

$$S \Rightarrow ABE \Rightarrow AaBE \Rightarrow aaBE$$
$$\Rightarrow aabSDE \Rightarrow aabbBdDE \Rightarrow$$
$$\Rightarrow aaabbccdDE \Rightarrow aabbccddE$$
$$\Rightarrow aabbccddeE \Rightarrow aabbccddee$$

Derivation Tree

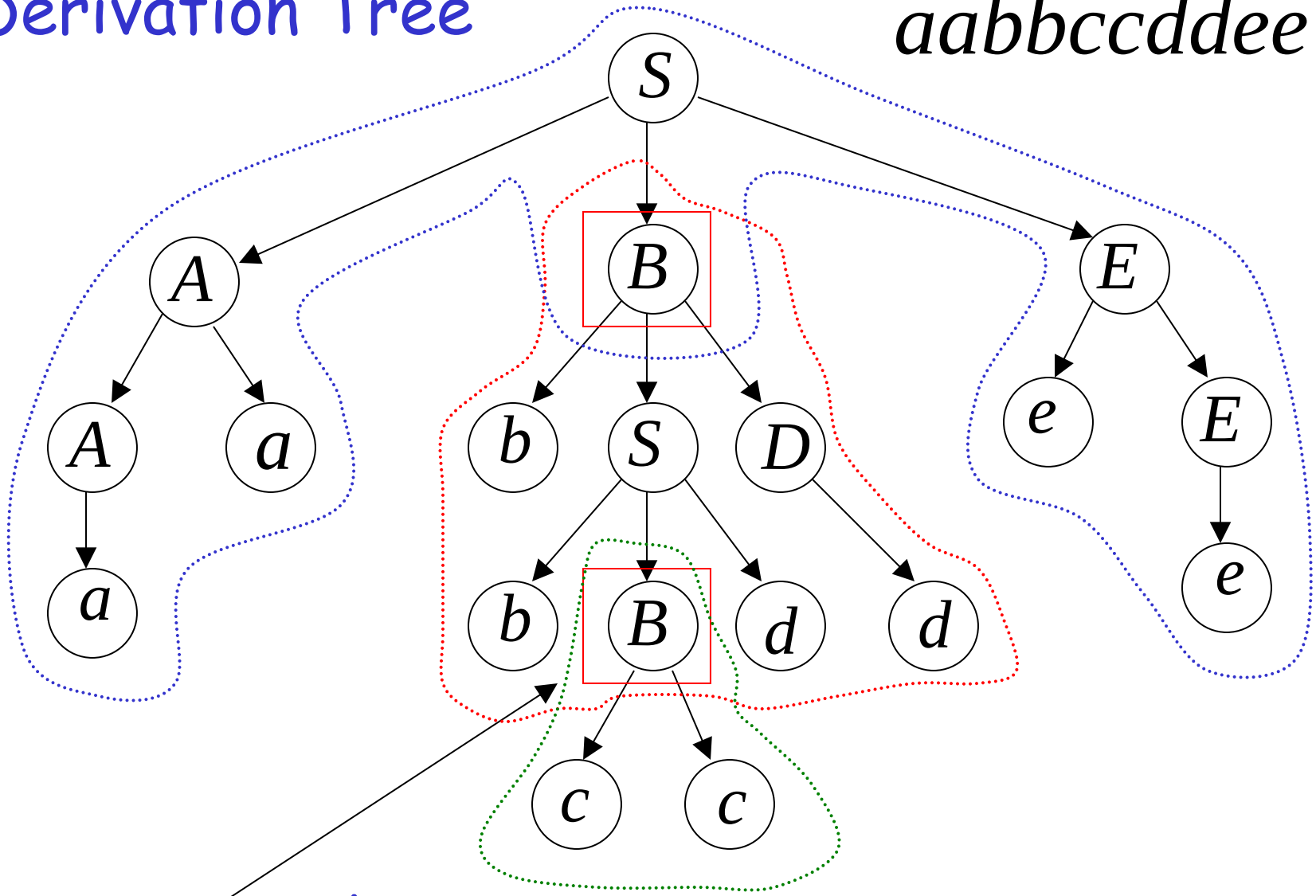
aabbccddee



Repeated variable

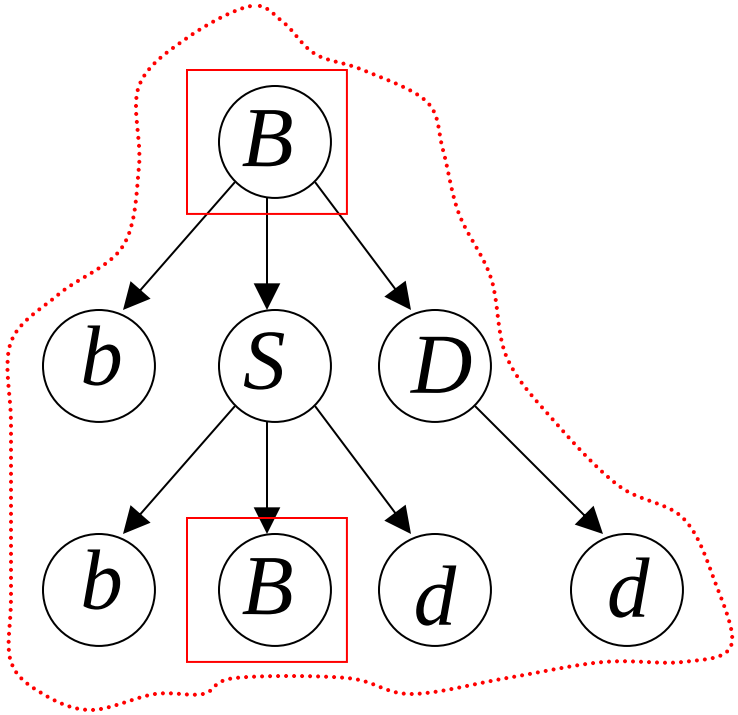
Derivation Tree

aabbccddee



Repeated variable

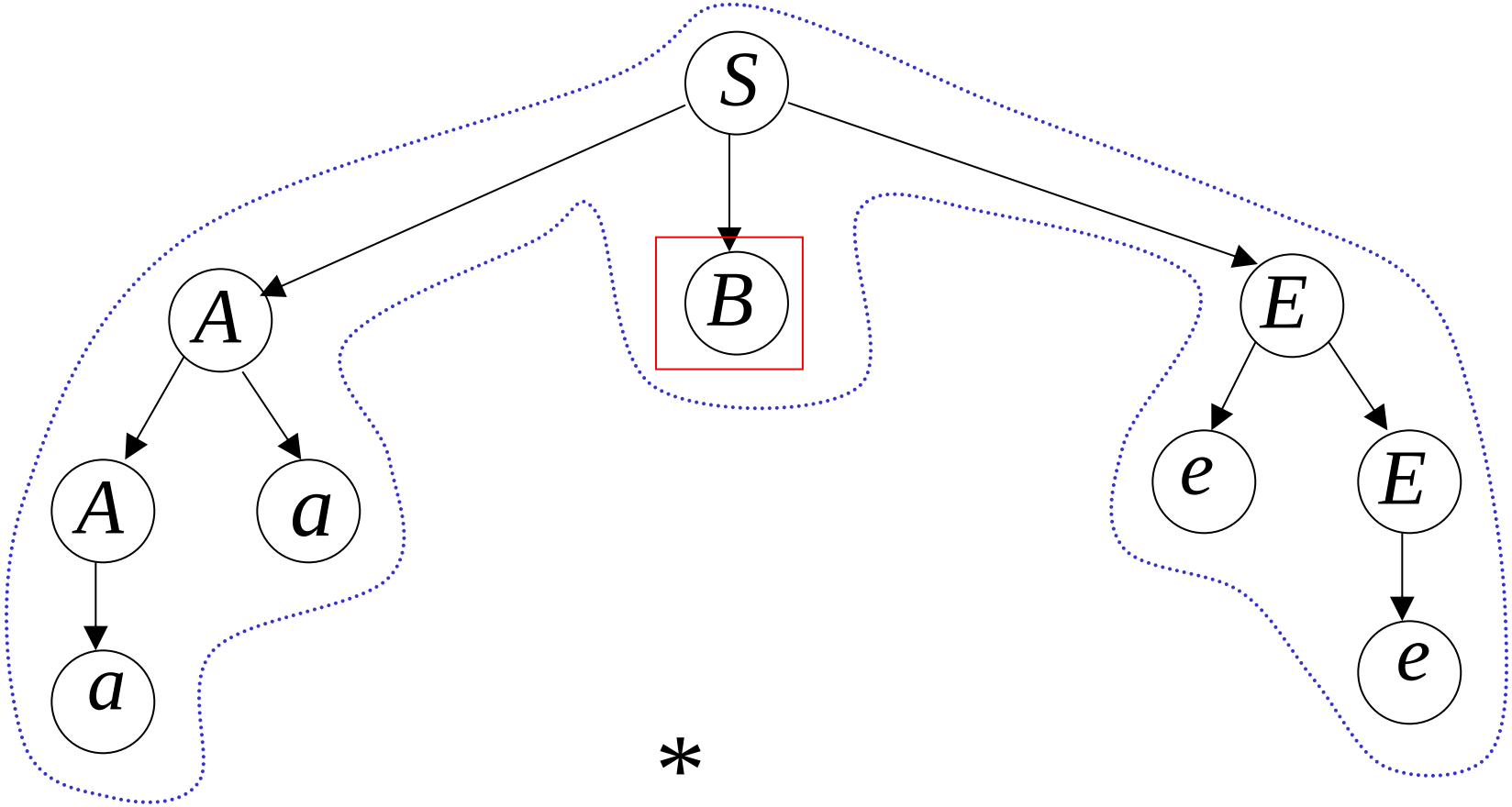
$$B \Rightarrow bSD \Rightarrow bbBdD \Rightarrow bbBdd$$



*

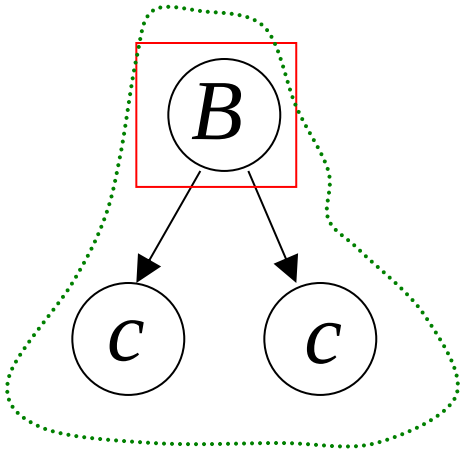
$$B \Rightarrow bbBdd$$

$$S \Rightarrow ABE \Rightarrow AaBE \Rightarrow aaBE \Rightarrow aaBeE \Rightarrow aaBee$$



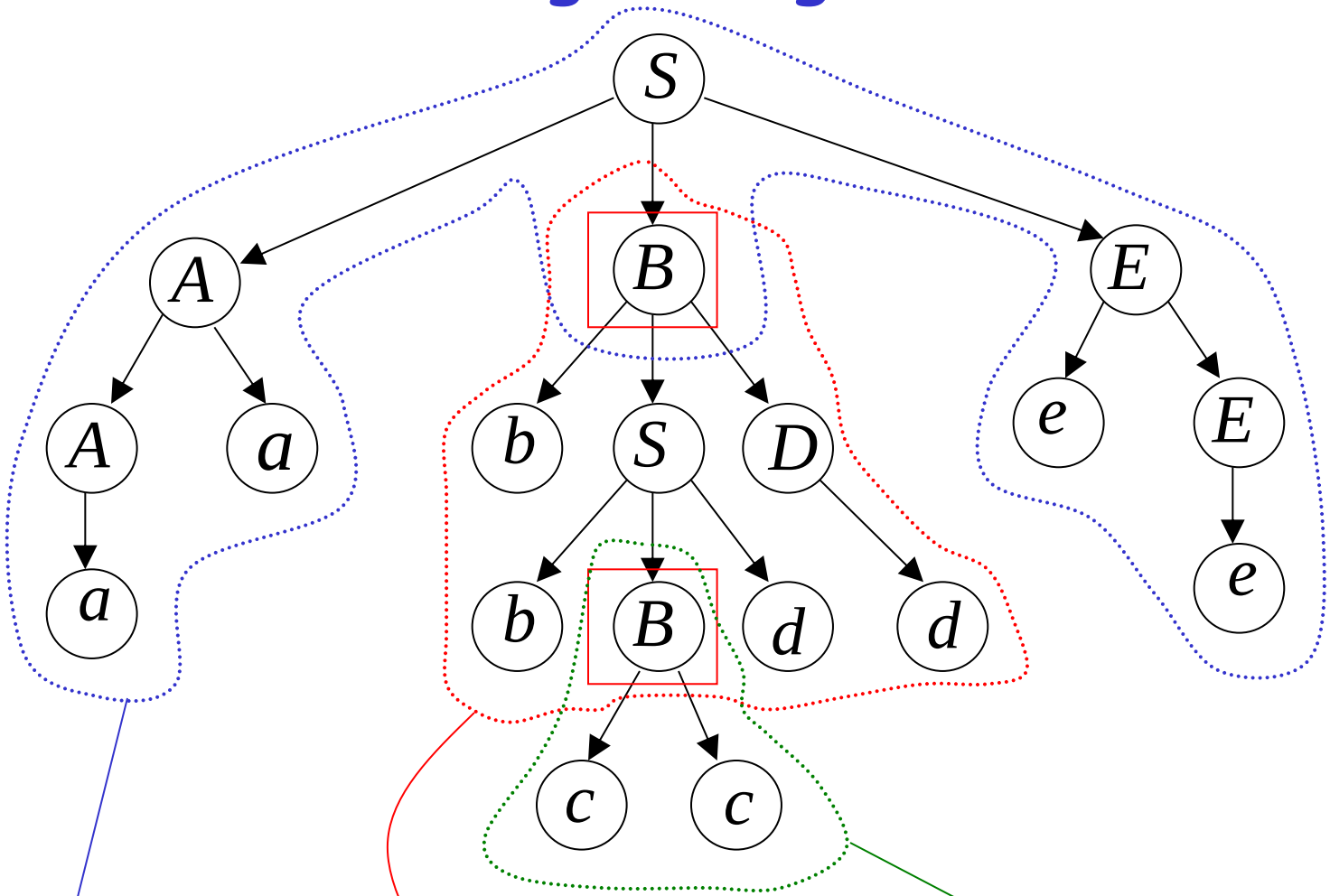
$$*$$

$$S \Rightarrow aaBee$$



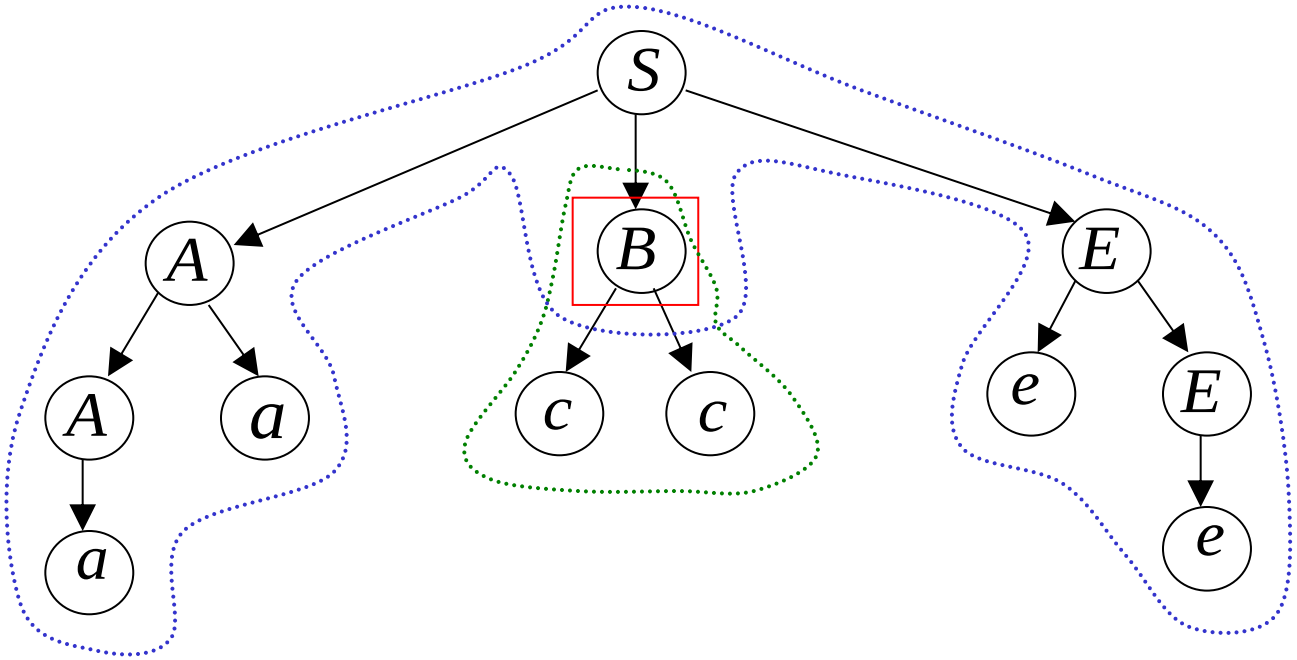
$$B \Rightarrow cc$$

Putting all together



$S \overset{*}{\Rightarrow} aaBee$ $\quad B \overset{*}{\Rightarrow} bbBdd$ $\quad B \Rightarrow cc$

We can remove the middle part



$$S \overset{*}{\Rightarrow} aa(bb)^0 cc(dd)^0 ee$$

$$S \overset{*}{\Rightarrow} aaBee \qquad B \overset{*}{\Rightarrow} bbBdd \qquad B \Rightarrow cc$$

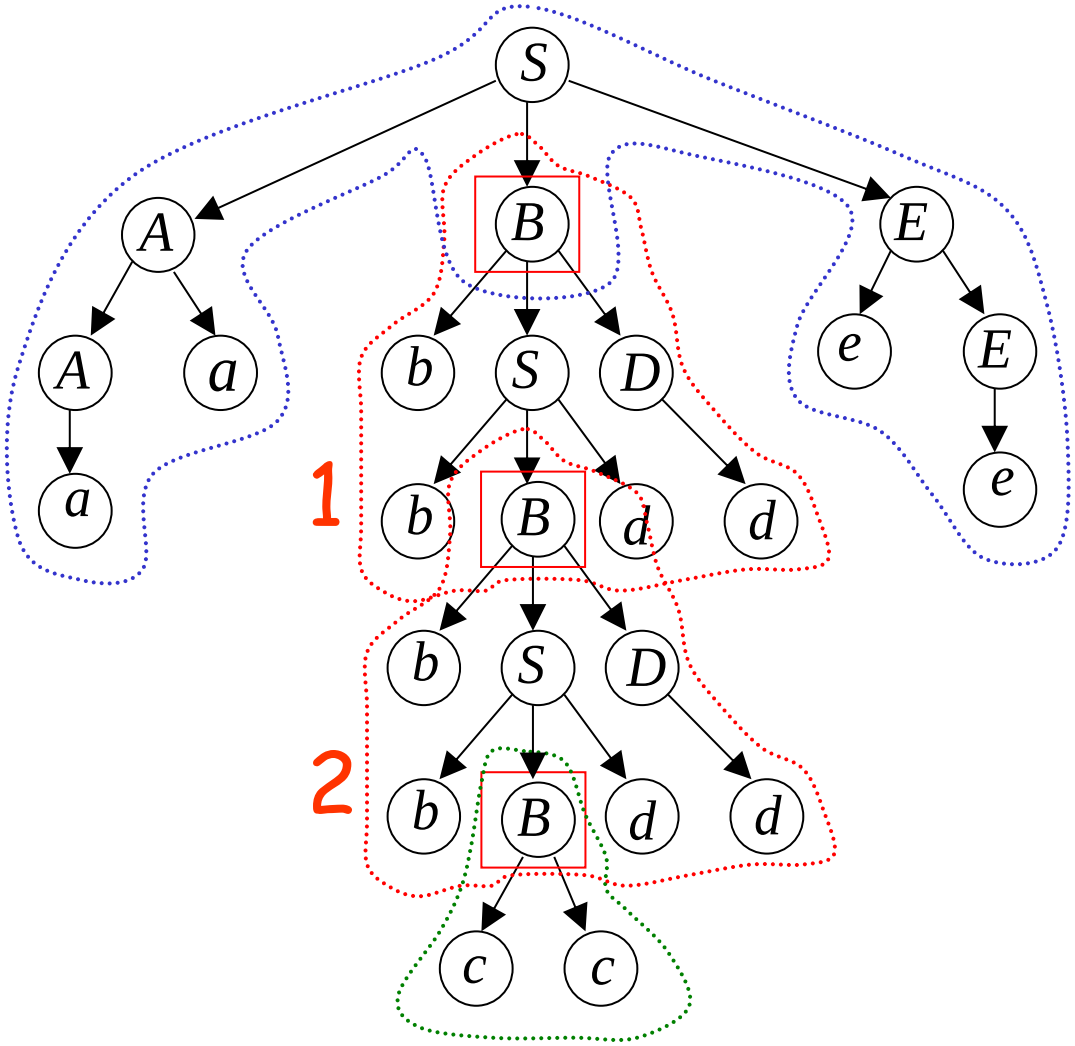


$$S \overset{*}{\Rightarrow} aaBee \overset{*}{\Rightarrow} aaccee \quad = aa(bb)^0 cc(dd)^0 ee$$



$$aa(bb)^0 cc(dd)^0 ee \quad \in L(G)$$

We can repeated middle part two times



$$S \overset{*}{\Rightarrow} aa(bb)^2cc(dd)^2ee$$

$$S \overset{*}{\Rightarrow} aaBee \qquad B \overset{*}{\Rightarrow} bbBdd \qquad B \Rightarrow cc$$



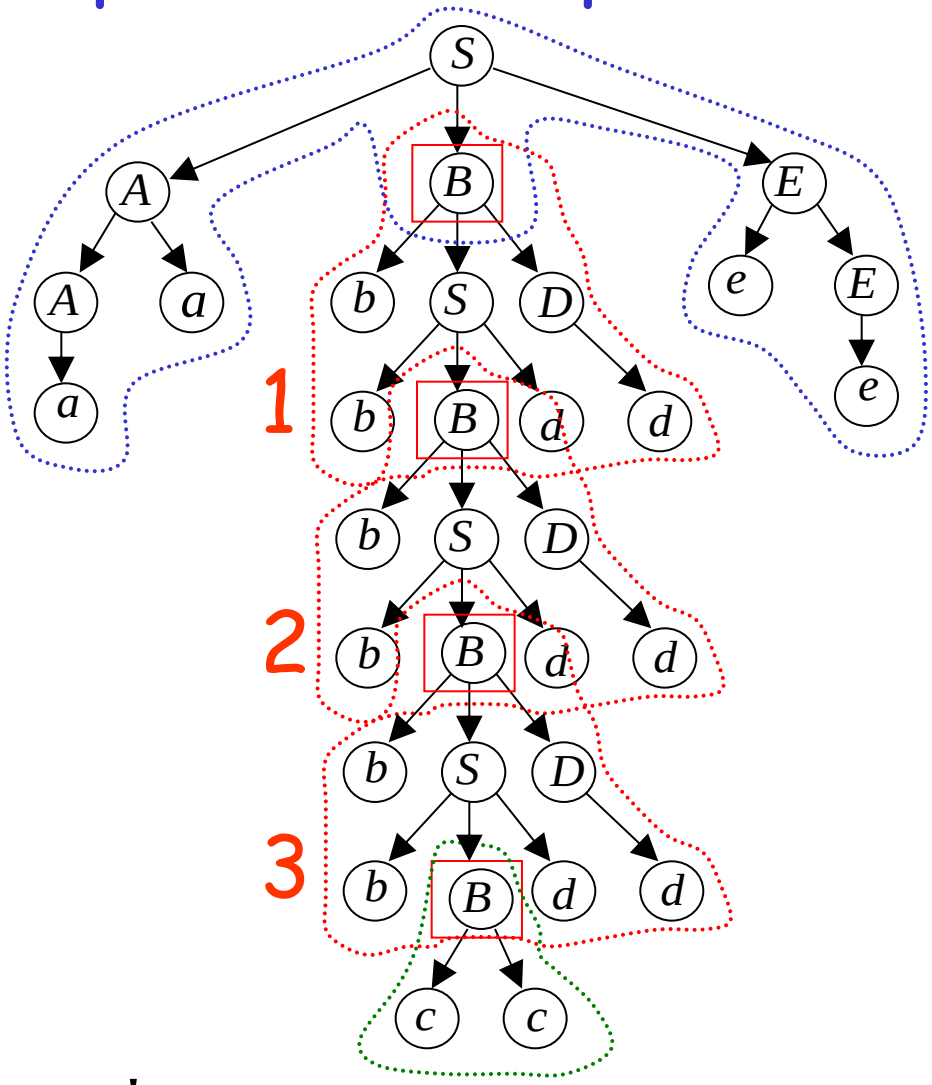
$$S \overset{*}{\Rightarrow} aaBee \overset{*}{\Rightarrow} aabbBddee$$

$$\overset{*}{\Rightarrow} aa(bb)^2 B(dd)^2 ee \overset{*}{\Rightarrow} aa(bb)^2 cc(dd)^2 ee$$



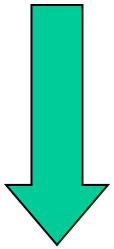
$$aa(bb)^2 cc(dd)^2 ee \in L(G)$$

We can repeat middle part three times



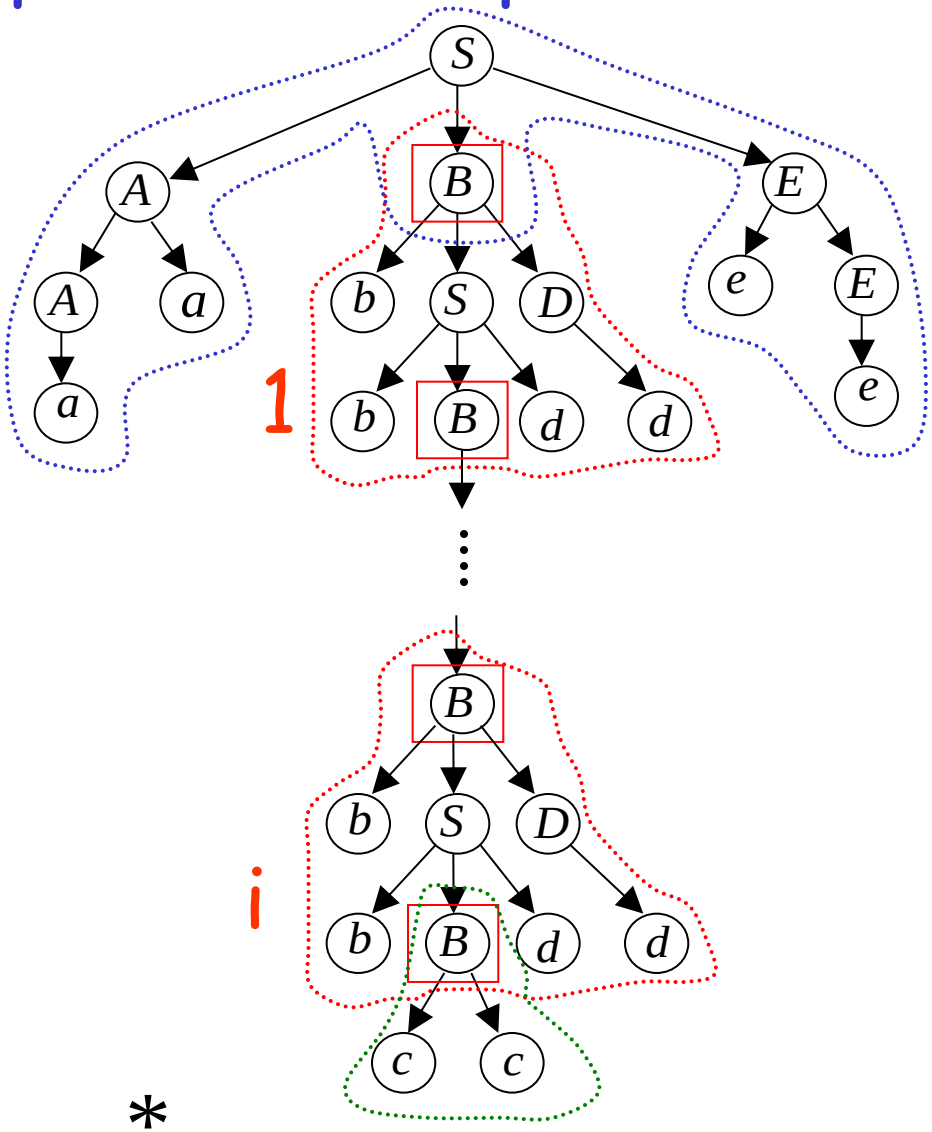
$$S \Rightarrow^{*} aa(bb)^3 cc(dd)^3 ee$$

$S \overset{*}{\Rightarrow} aaBee$ $\quad B \overset{*}{\Rightarrow} bbBdd$ $\quad B \Rightarrow cc$



$$S \overset{*}{\Rightarrow} aa(bb)^3 cc(dd)^3 ee \quad \in L(G)$$

Repeat middle part i times



$$S \Rightarrow^{*} aa(bb)^{i}cc(dd)^{i}ee$$

$$S \overset{*}{\Rightarrow} aaBee \qquad B \overset{*}{\Rightarrow} bbBdd \qquad B \Rightarrow cc$$



$$S \overset{*}{\Rightarrow} aa(bb)^i cc(dd)^i ee \quad \in L(G)$$

For any $i \geq 0$

From Grammar

$$S \to ABE \mid bBd$$

$$A \to Aa \mid a$$

$$B \to bSD \mid cc$$

$$D \to Dd \mid d$$

$$E \to eE \mid e$$

and given string

$$aabbccddee \quad \in L(G)$$

We inferred that a family of strings is in $L(G)$

$$S \overset{*}{\Rightarrow} aa(bb)^i cc(dd)^i ee \quad \in L(G)$$ for any $i \geq 0$

Arbitrary Grammars

Consider now an arbitrary infinite context-free language L

Let G be the grammar of $L - \{\varepsilon\}$

Take G so that it has no unit-productions and no ε-productions

(remove them)

Let r be the number of variables

Let t be the maximum right-hand size of any production

Example:

$$S \to ABE \mid bBd$$
$$A \to Aa \mid a$$
$$B \to bSD \mid cc$$
$$D \to Dd \mid d$$
$$E \to eE \mid e$$

$r = 5$ (variables)

$t = 3$

Claim:

Take string $w \in L(G)$ with $|w| > t^r$.

Then in the derivation tree of w

there is a path from the root to a leaf

where a variable of G is repeated

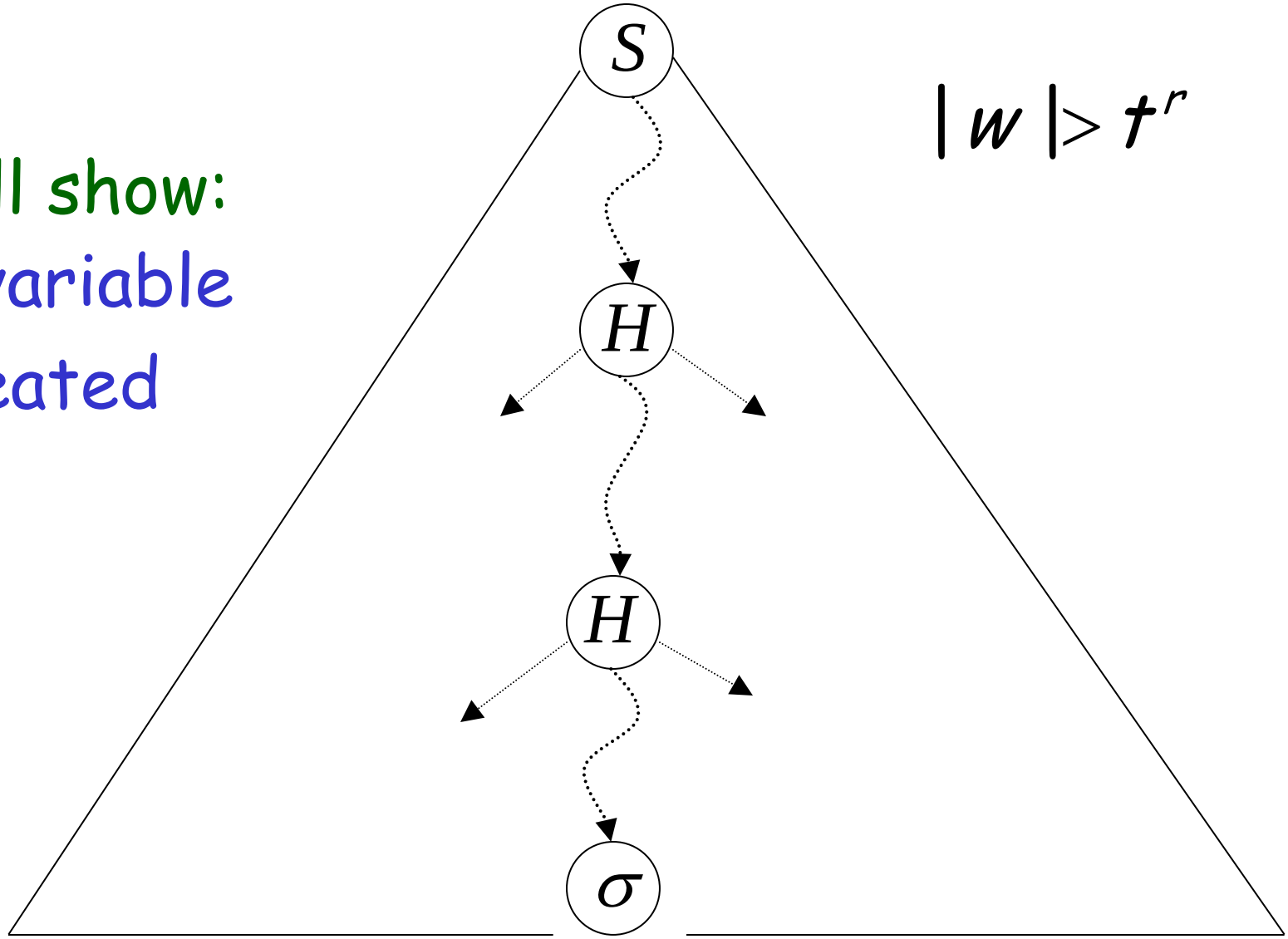
Proof:

Proof by contradiction

Derivation tree of w

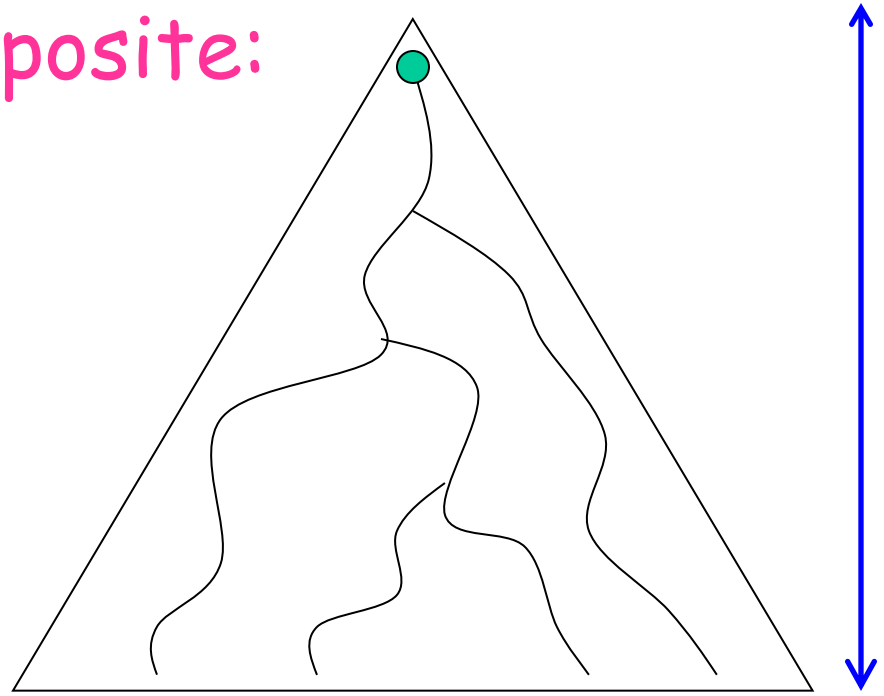
$|w| > t^r$

We will show:
some variable
is repeated



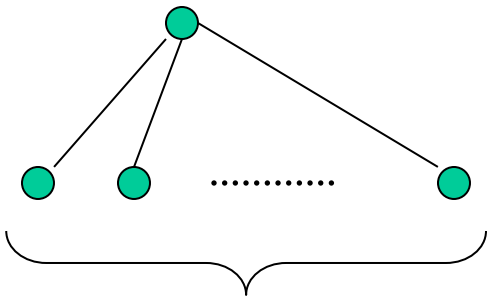
First we show that the tree of w has at least $r+2$ levels of nodes

Suppose the opposite:



At most $r+1$ Levels

Maximum number of nodes per level



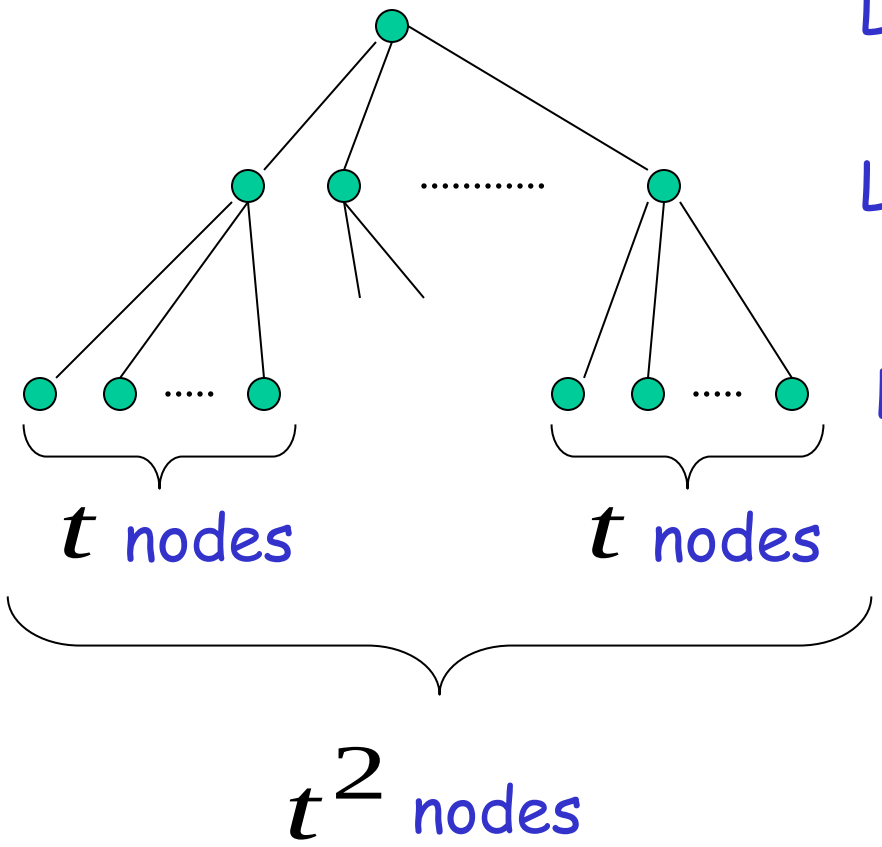
t nodes

Level 0: 1 nodes

Level 1: t nodes

The maximum right-hand side of any production

Maximum number of nodes per level

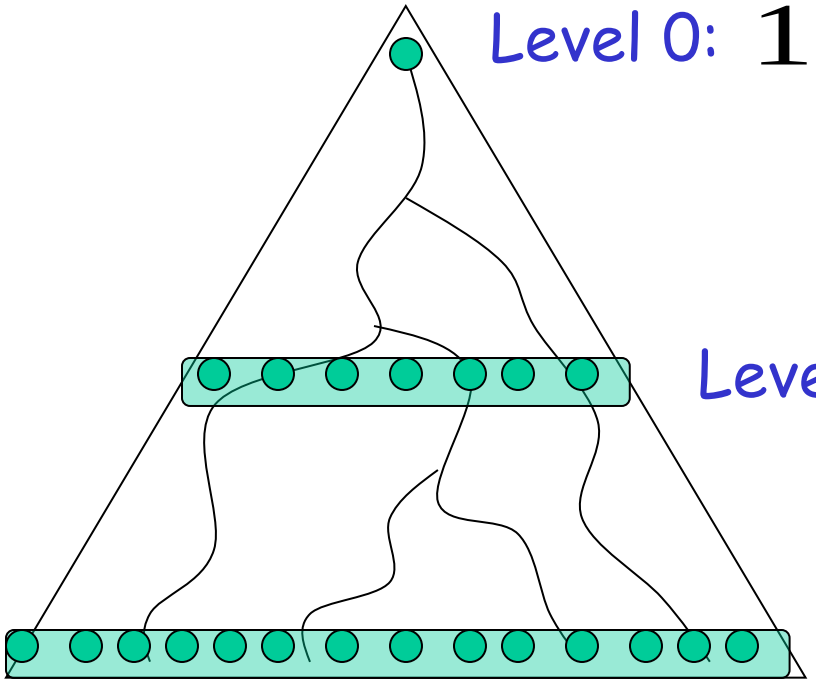


Level 0: 1 nodes

Level 1: t nodes

Level 2: t^2 nodes

Maximum number of nodes per level



At most $r+1$ Levels

Maximum possible string length
= max nodes at level r = $\boxed{t^r}$

Therefore,

maximum length of string w : $|w| \leq t^r$

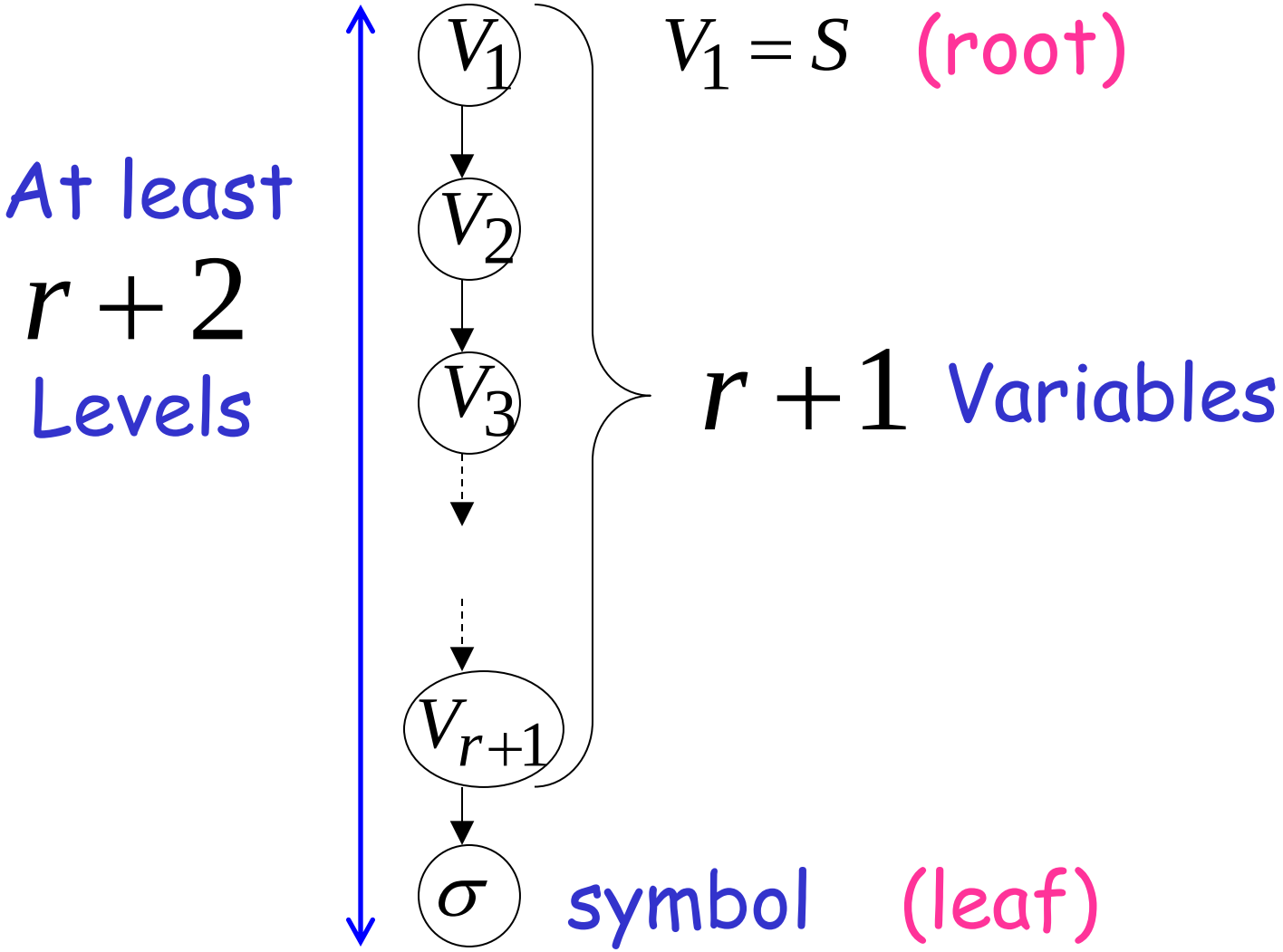
However we took, $|w| > t^r$

Contradiction!!!

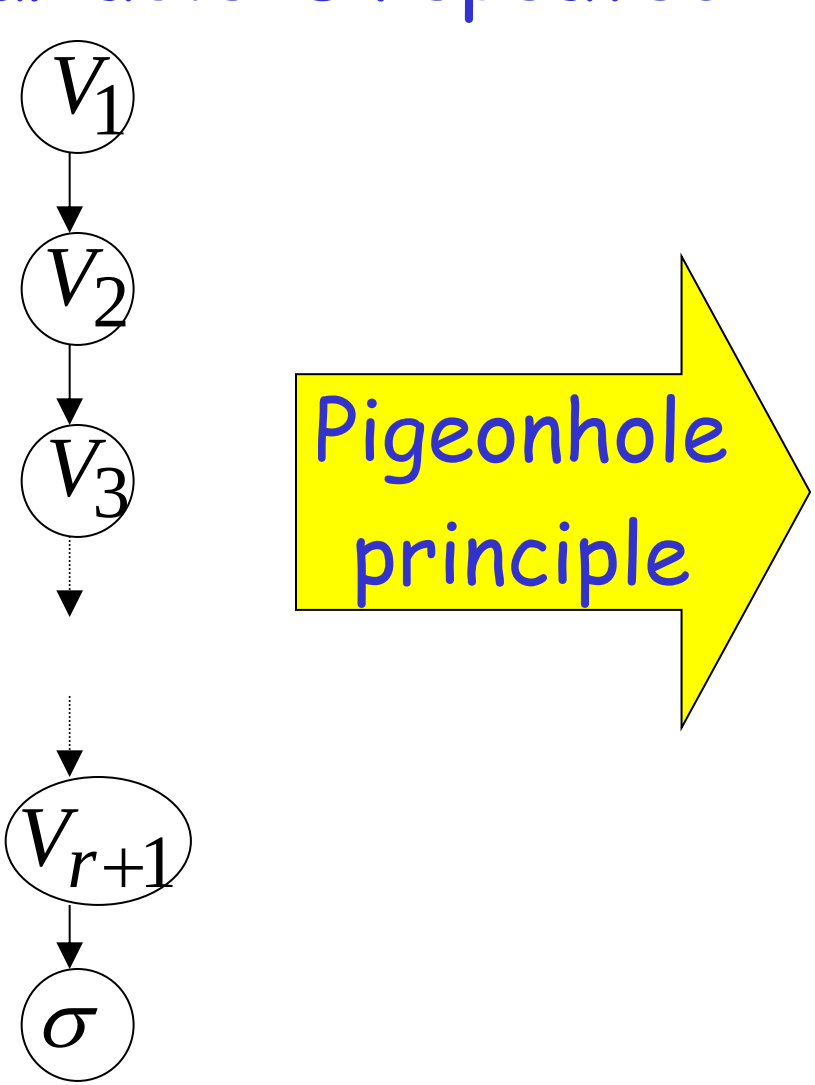
Therefore,

the tree must have at least $r+2$ levels

Thus, there is a path from the root to a leaf with at least $r+2$ nodes



Since there are at most r different variables, some variable is repeated

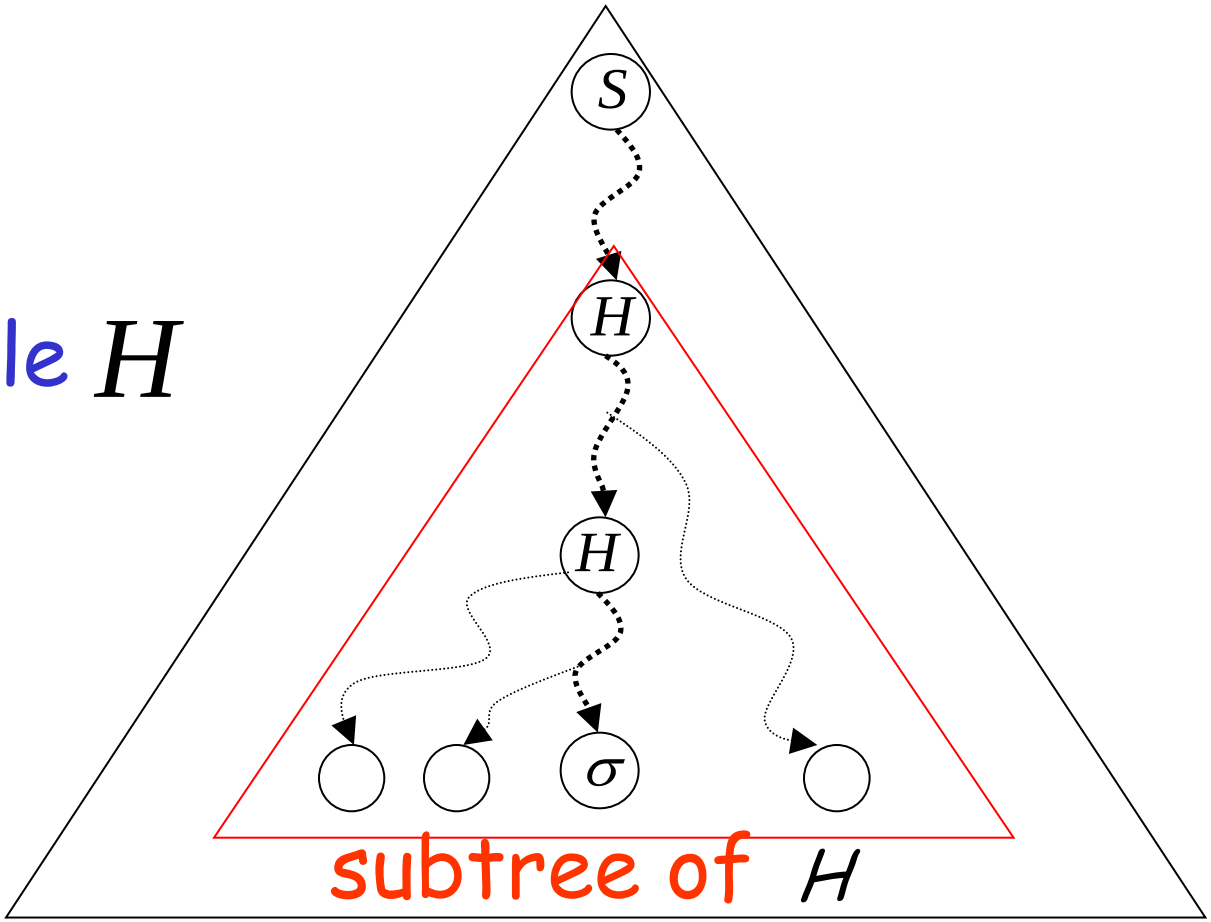


END OF CLAIM PROOF

Take now a string w with $|w| > t^r$

From claim:

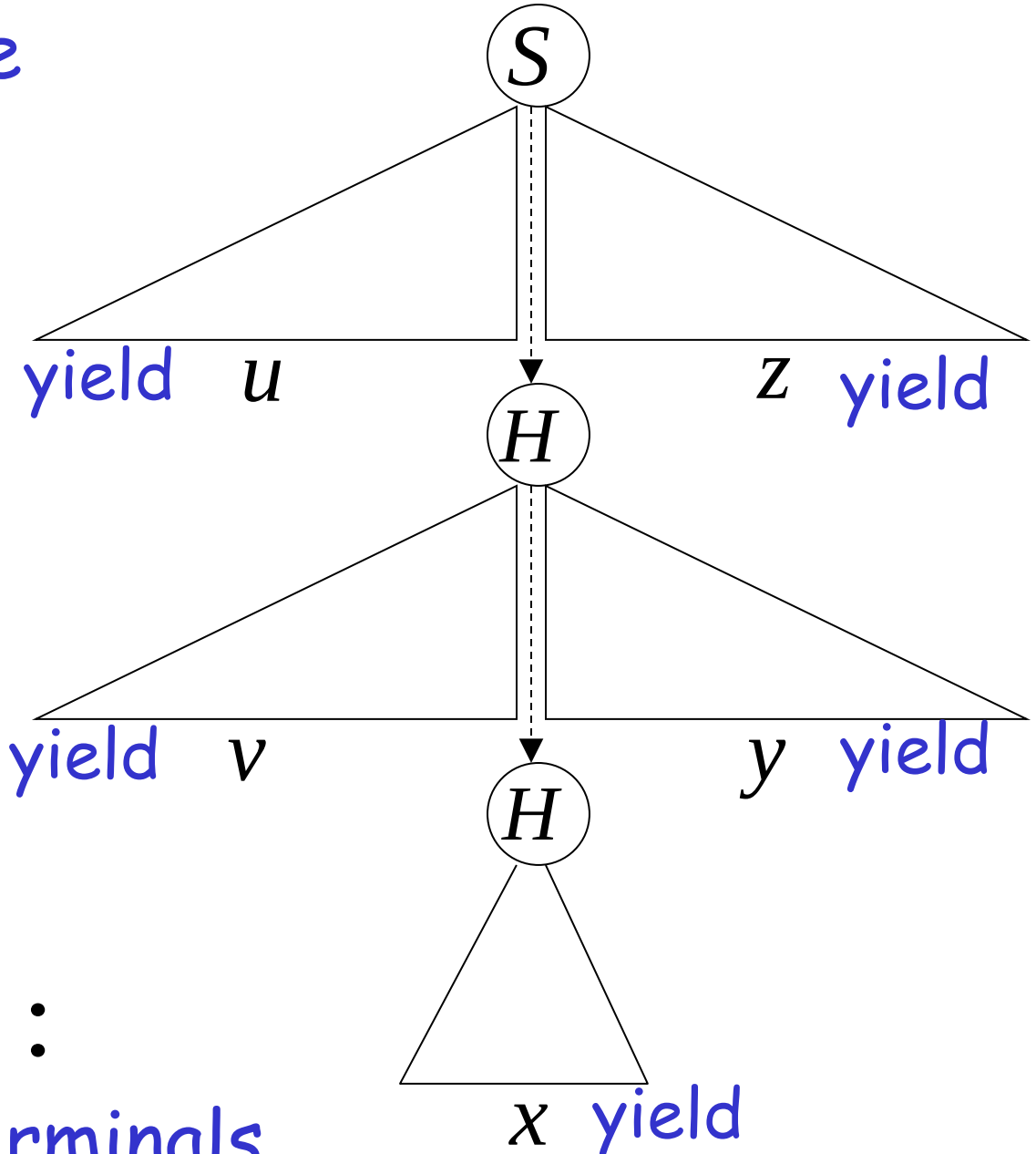
some variable H is repeated



Take H to be the deepest, so that only H is repeated in subtree

We can write

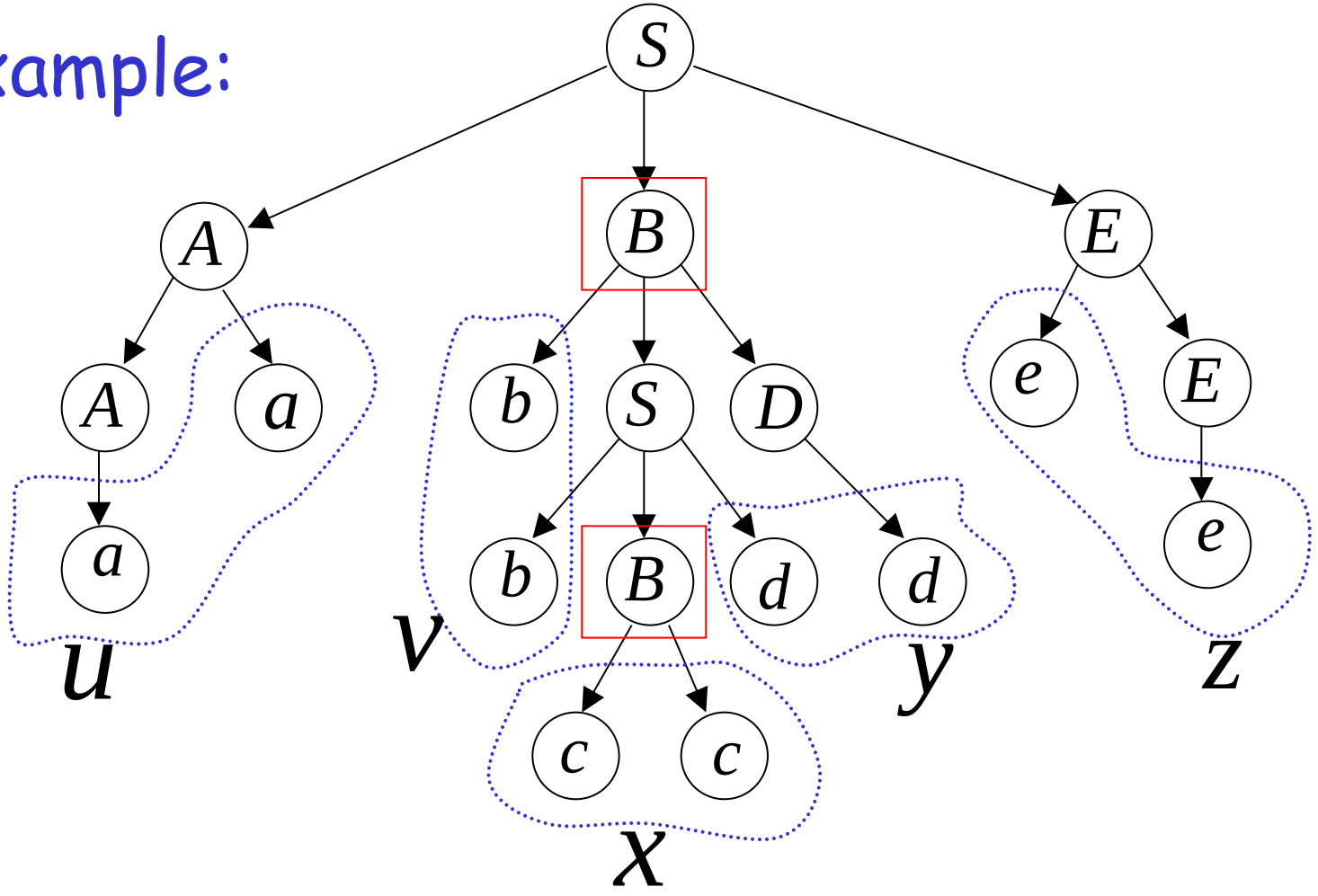
$w = uvxyz$



u, v, x, y, z :

Strings of terminals

Example:



$u = aa$

$v = bb$

$x = cc$

$y = dd$

$z = ee$

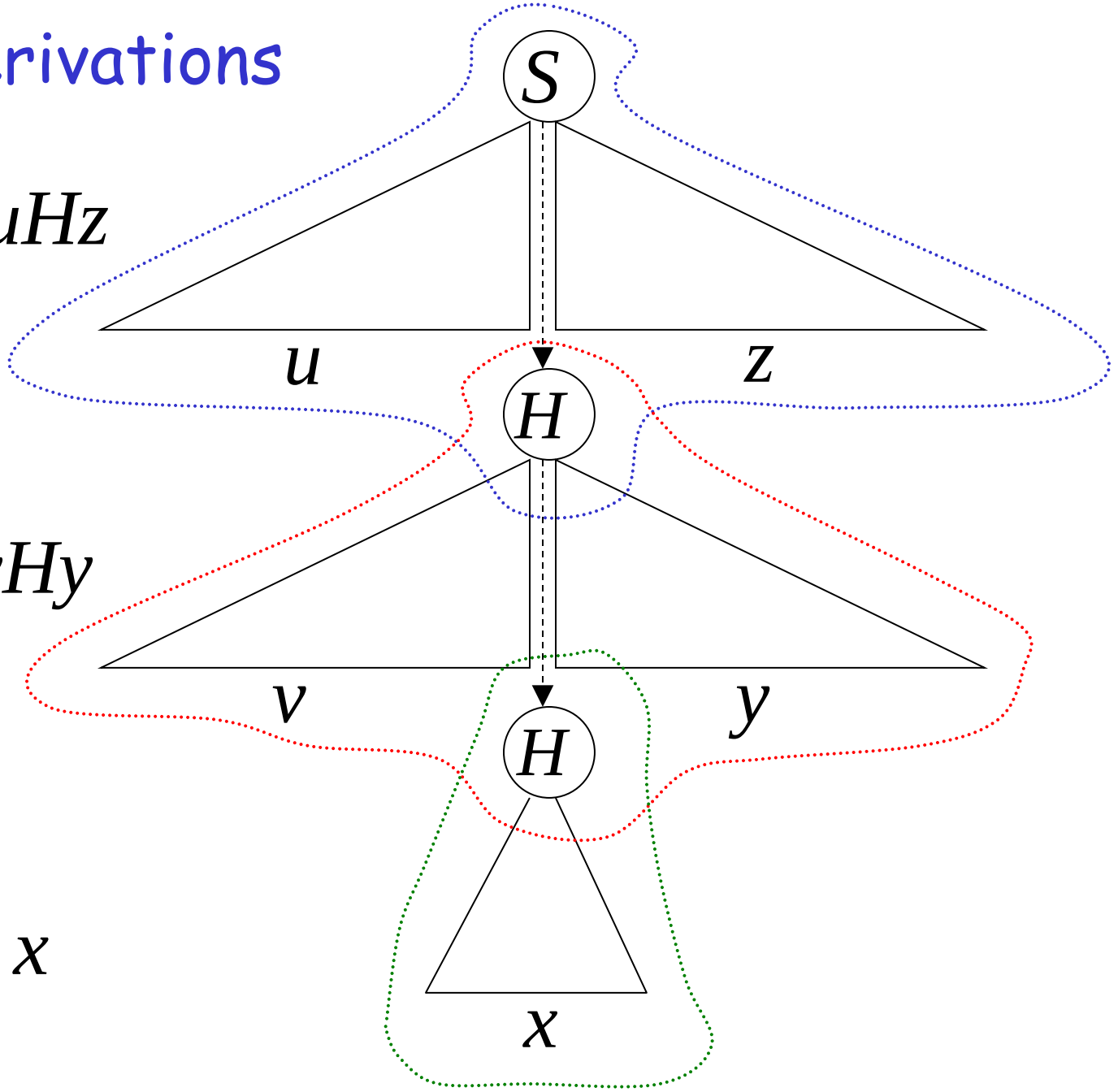
B corresponds to H

Possible derivations

$$S \overset{*}{\Rightarrow} uHz$$

$$H \overset{*}{\Rightarrow} vHy$$

$$H \overset{*}{\Rightarrow} x$$



Example:

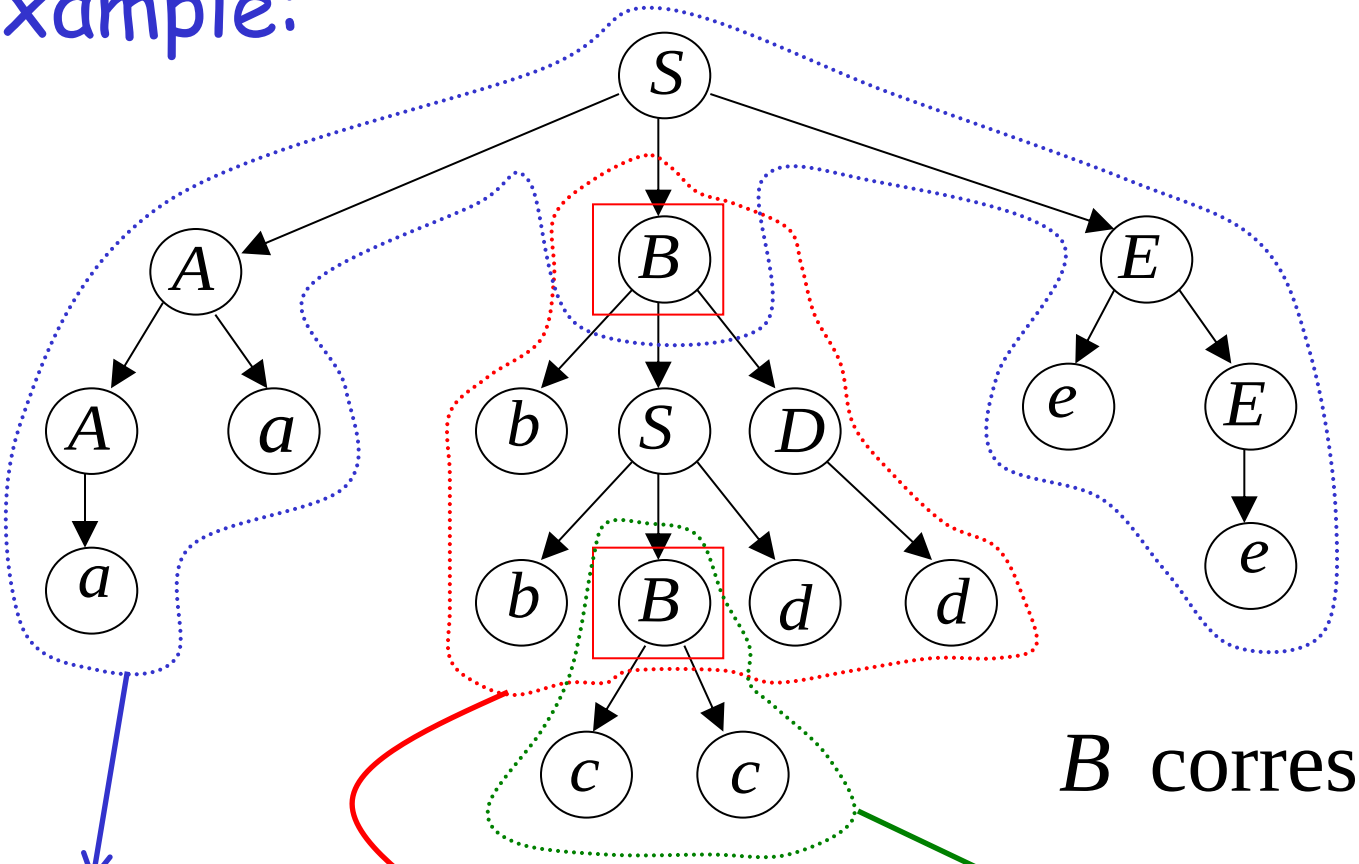
$u = aa$

$v = bb$

$x = cc$

$y = dd$

$z = ee$



B corresponds to H

$S \overset{*}{\Rightarrow} uHz$

$H \overset{*}{\Rightarrow} vHy$

$H \overset{*}{\Rightarrow} x$

$S \overset{*}{\Rightarrow} aaBee$

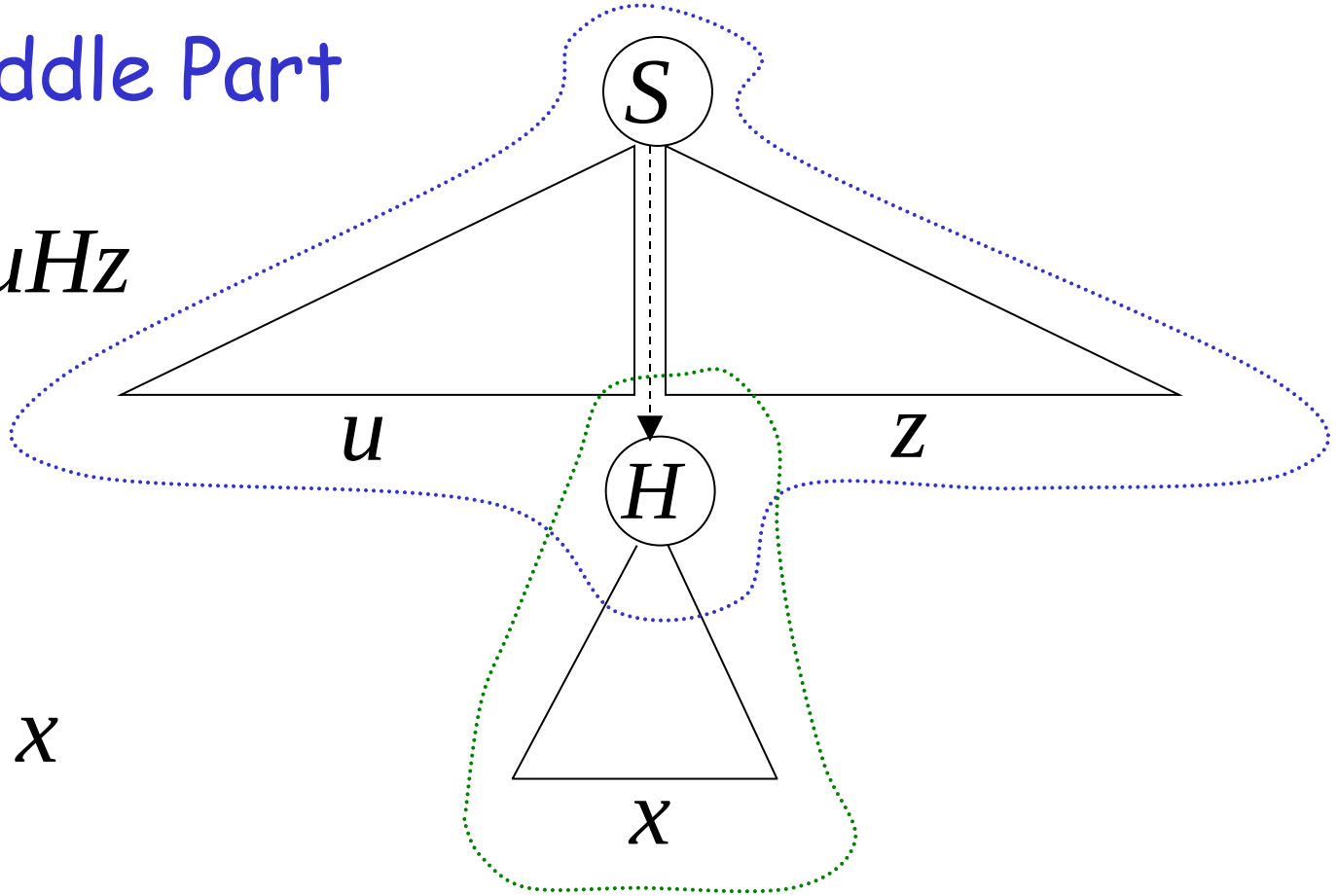
$B \overset{*}{\Rightarrow} bbBdd$

$B \Rightarrow cc$

Remove Middle Part

$$S \overset{*}{\Rightarrow} uHz$$

$$H \overset{*}{\Rightarrow} x$$



Yield: $uxz = uv^0xy^0z$

$$S \overset{*}{\Rightarrow} uHz \overset{*}{\Rightarrow} uxz = uv^0xy^0z \in L(G)$$

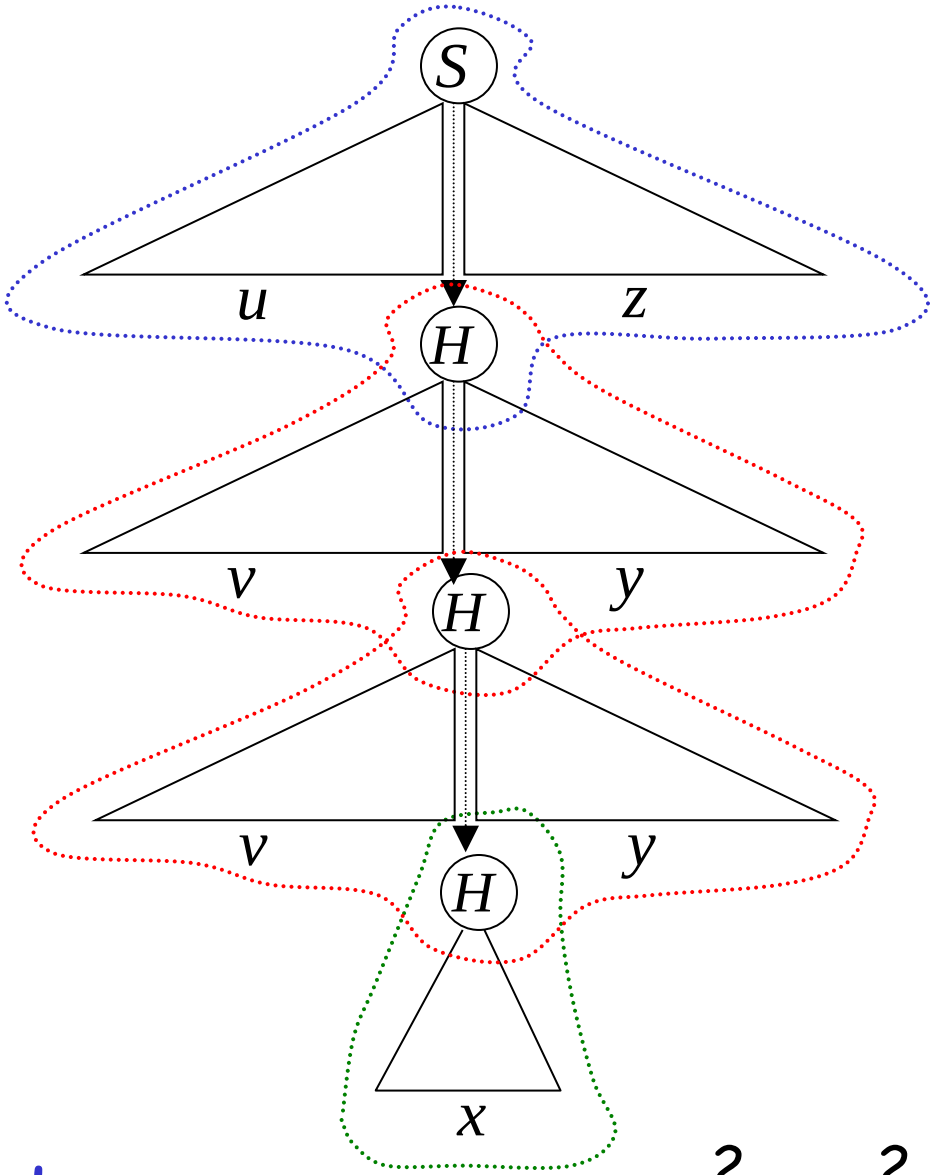
Repeat Middle part two times

$$S \overset{*}{\Rightarrow} uHz$$

$$H \overset{*}{\Rightarrow} vHy$$

$$H \overset{*}{\Rightarrow} vHy$$

$$H \overset{*}{\Rightarrow} x$$



Yield: $uvvxyyz = uv^2xy^2z$

$$S \overset{*}{\Rightarrow} uHz \qquad H \overset{*}{\Rightarrow} vHy \qquad H \overset{*}{\Rightarrow} x$$



$$S \overset{*}{\Rightarrow} uHz \overset{*}{\Rightarrow} uvHyz \overset{*}{\Rightarrow} uvvHyyz$$

$$\overset{*}{\Rightarrow} uvvxyyz = uv^2xy^2z \in L(G)$$

Repeat Middle part i times

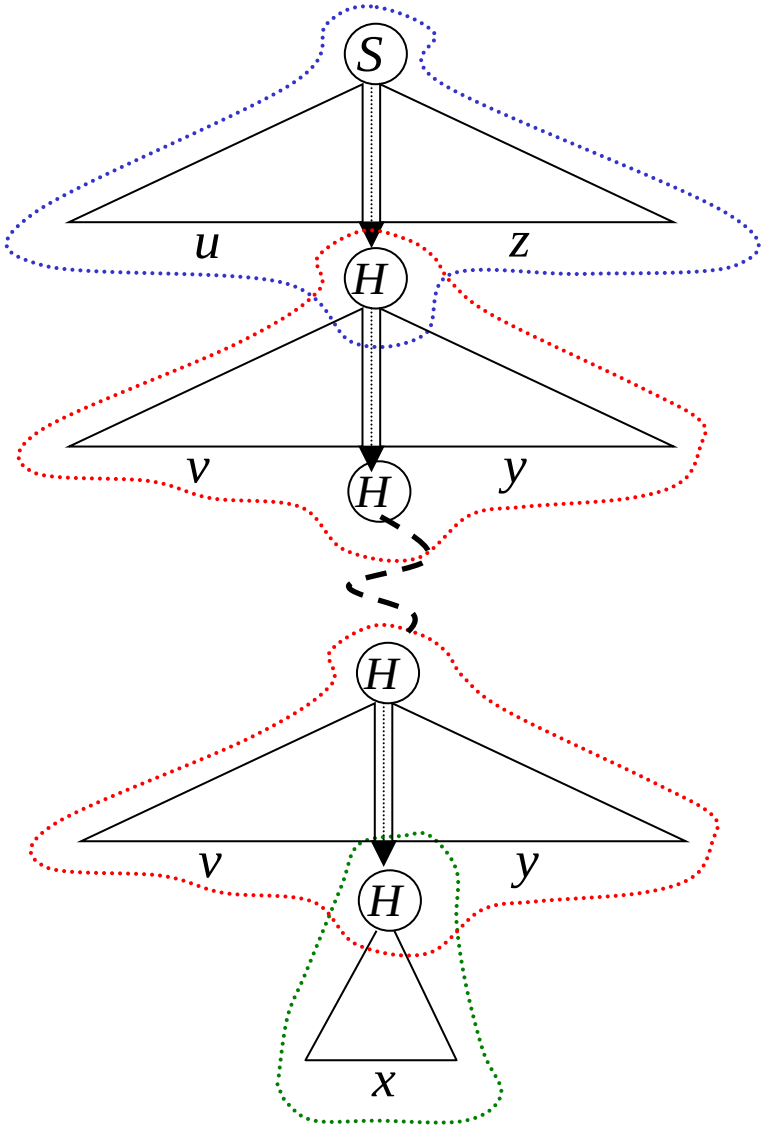
$$S \overset{*}{\Rightarrow} uHz$$

$$H \overset{*}{\Rightarrow} vHy$$

$$\vdots$$

$$H \overset{*}{\Rightarrow} vHy$$

$$H \overset{*}{\Rightarrow} x$$



Yield: $uv^i xy^i z$

$$S \overset{*}{\Rightarrow} uHz \qquad H \overset{*}{\Rightarrow} vHy \qquad H \overset{*}{\Rightarrow} x$$



$$S \overset{*}{\Rightarrow} uHz \overset{*}{\Rightarrow} uvHyz \overset{*}{\Rightarrow} uvvHyyz \overset{*}{\Rightarrow}$$

$$\overset{*}{\Rightarrow} \ldots$$

$$\overset{*}{\Rightarrow} uv^{i}Hy^{i}z \overset{*}{\Rightarrow} uv^{i}xy^{i}z \in L(G)$$

Therefore,

$$|w| > t^r$$

If we know that: $w = uvxyz \in L(G)$

then we also know: $uv^i xy^i z \in L(G)$

For all $i \geq 0$

since

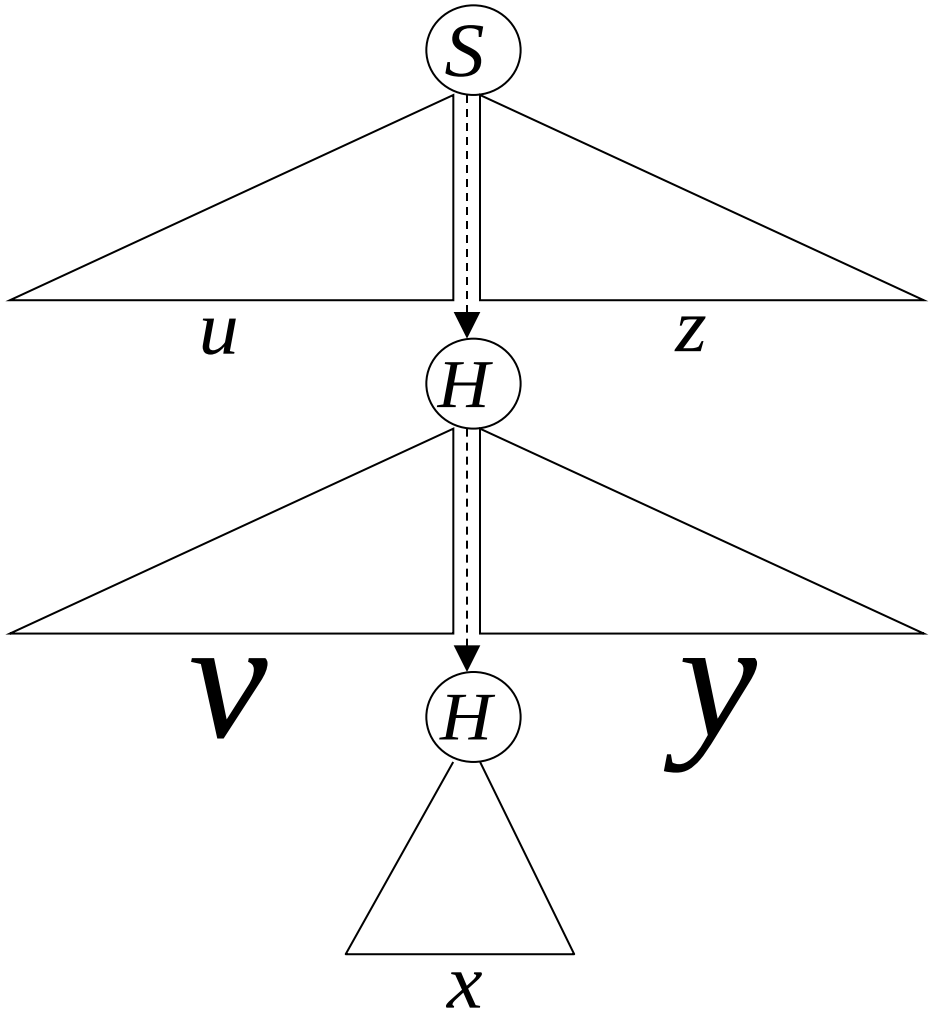
$L(G) = L - \{\lambda\}$

$$uv^i xy^i z \in L$$

Observation 1:

$$|vy| \geq 1$$

Since G has no unit and ε-productions

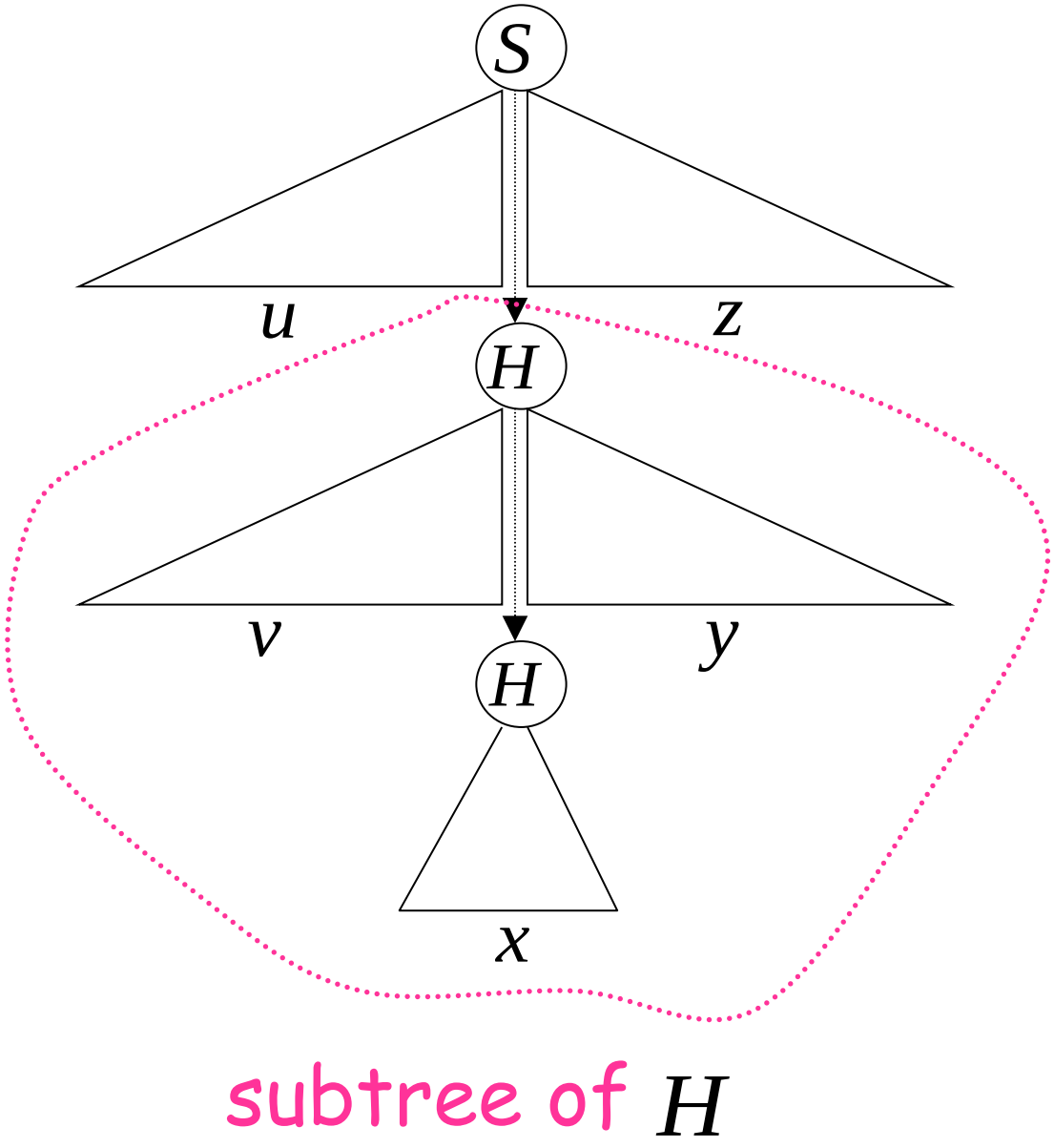


At least one of v or y is not ε

Observation 2:

$|vxy| \leq t^{r+1}$

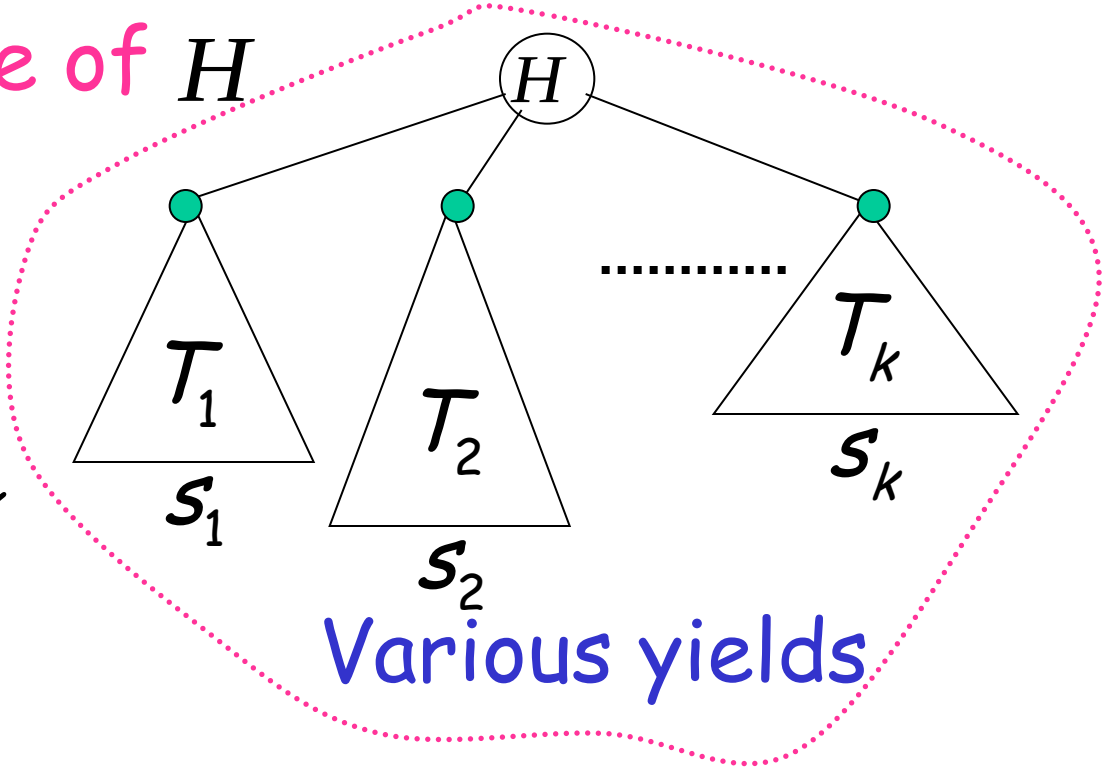
since in subtree only variable H is repeated



subtree of H

Explanation follows....

subtree of H



$vxy = s_1 s_2 \cdots s_k$

Various yields

$|s_j| \leq t^r$ since no variable is repeated in T_j

$$|vxy| = \sum_{j=1}^{k} |s_j| \leq k \cdot t^r \leq t \cdot t^r = t^{r+1}$$

Maximum right-hand side of any production

Thus, if we choose critical length

$$p = t^{r+1} > t^{r}$$

then, we obtain the pumping lemma for context-free languages

The Pumping Lemma:

For any infinite context-free language L

there exists an integer p such that

for any string $w \in L, \quad |w| \geq p$

we can write $w = uvxyz$

with lengths $|vxy| \leq p$ and $|vy| \geq 1$

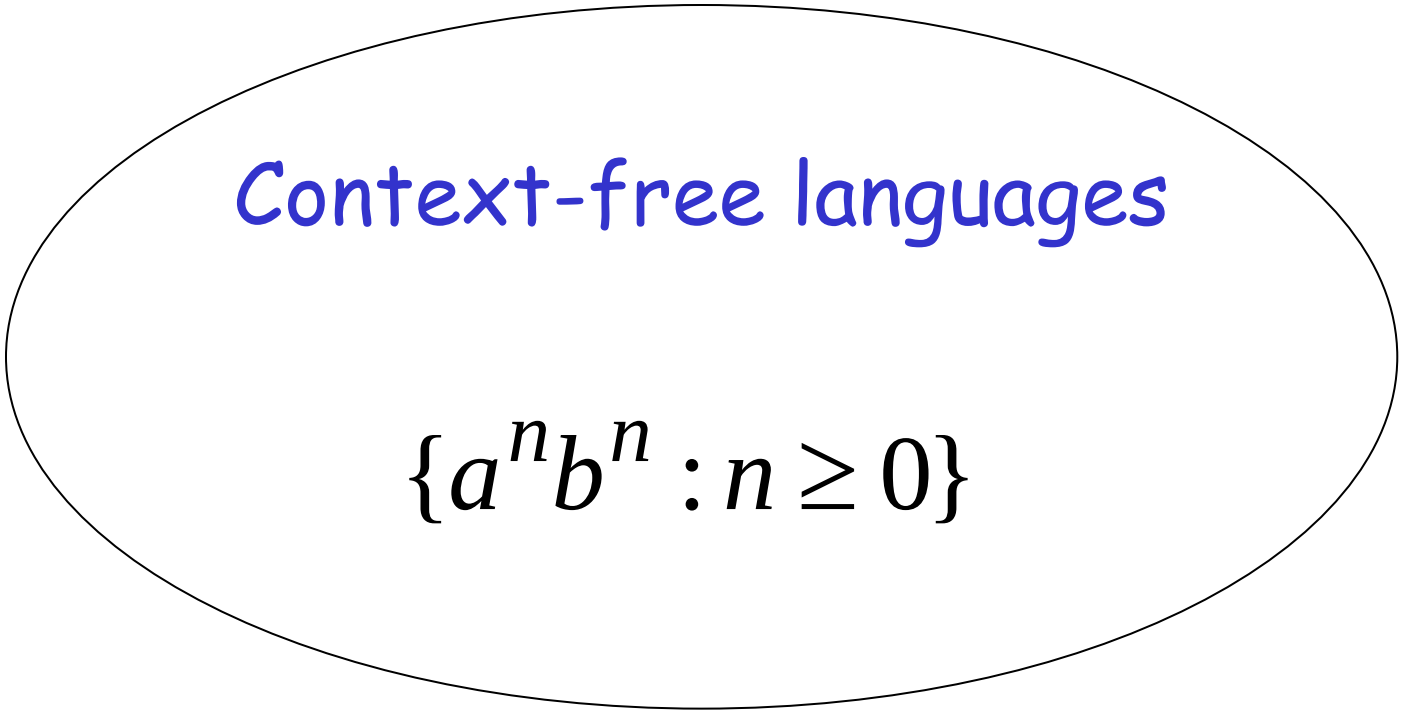
and it must be that:

$$uv^i xy^i z \in L, \quad \text{for all } i \geq 0$$

Applications of The Pumping Lemma

Non-context free languages

$$\{a^n b^n c^n : n \geq 0\}$$



Theorem: The language

$$L = \{a^n b^n c^n : n \geq 0\}$$

is **not** context free

Proof: Use the Pumping Lemma for context-free languages

$$L = \{a^n b^n c^n : n \geq 0\}$$

Assume for contradiction that L

is context-free

Since L is context-free and infinite

we can apply the pumping lemma

$$L = \{a^n b^n c^n : n \geq 0\}$$

Let p be the critical length of the pumping lemma

Pick any string $w \in L$ with length $|w| \geq p$

We pick: $w = a^p b^p c^p$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

From pumping lemma:

we can write: $w = uvxyz$

with lengths $|vxy| \leq p$ and $|vy| \geq 1$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Pumping Lemma says:

$$uv^i xy^i z \in L \quad \text{for all} \quad i \geq 0$$

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

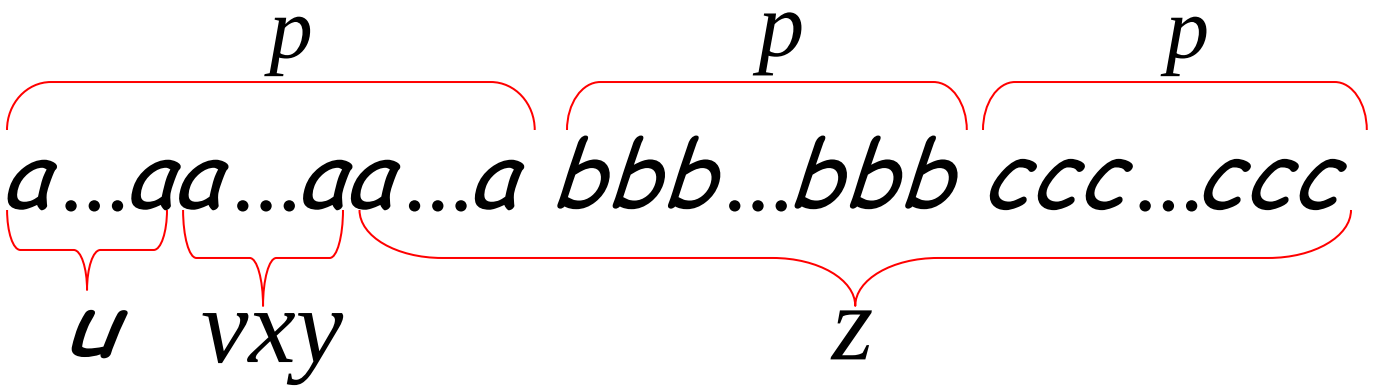
We examine **all** the possible locations of string vxy in w

$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 1: vxy is in a^p

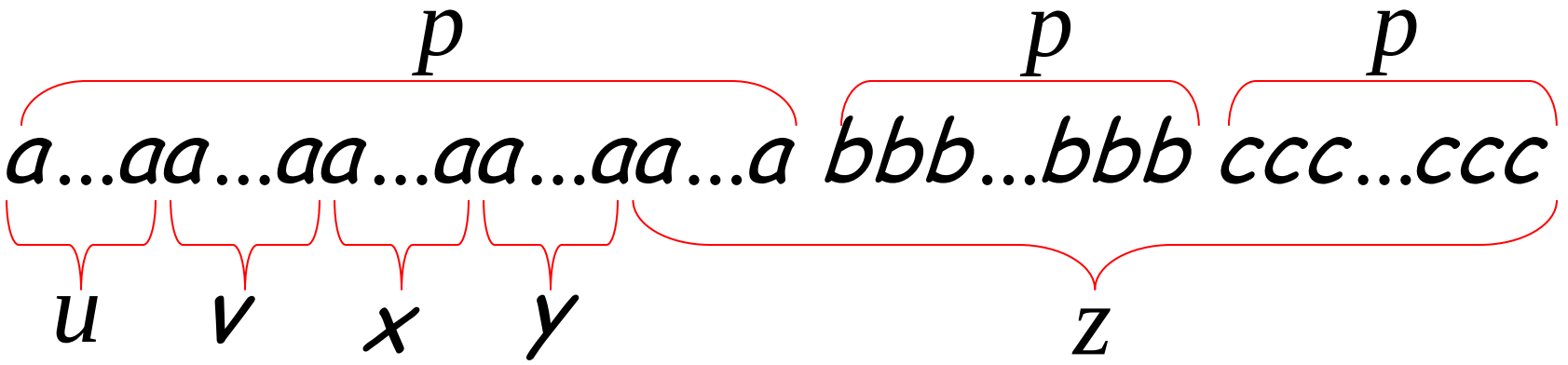


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

$$v = a^{k_1} \qquad y = a^{k_2} \qquad k_1 + k_2 \geq 1$$

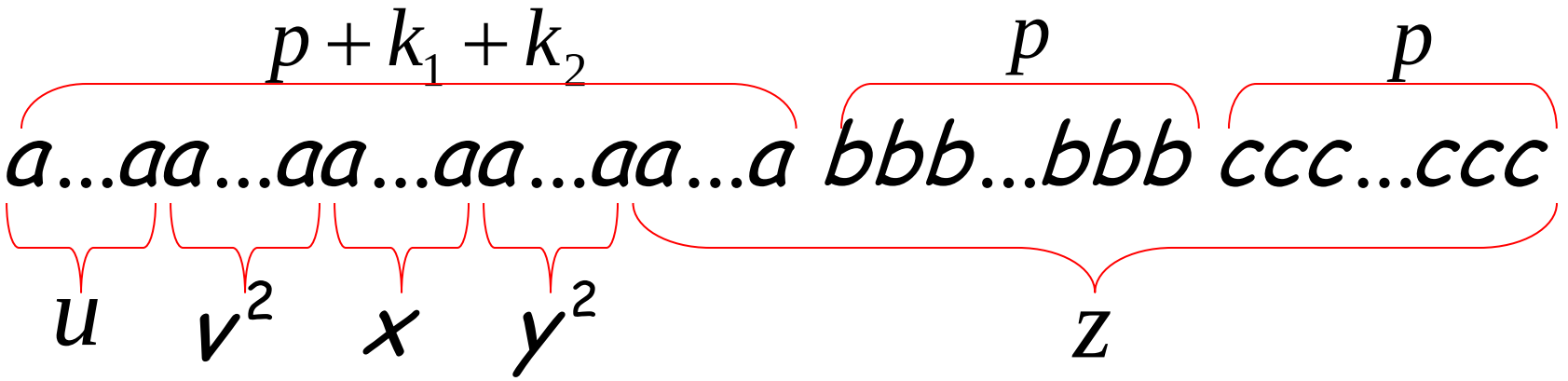


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

$$v = a^{k_1} \qquad y = a^{k_2} \qquad k_1 + k_2 \geq 1$$



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

However: $uv^2xy^2z = a^{p+k_1+k_2} b^p c^p \notin L$

Contradiction!!!

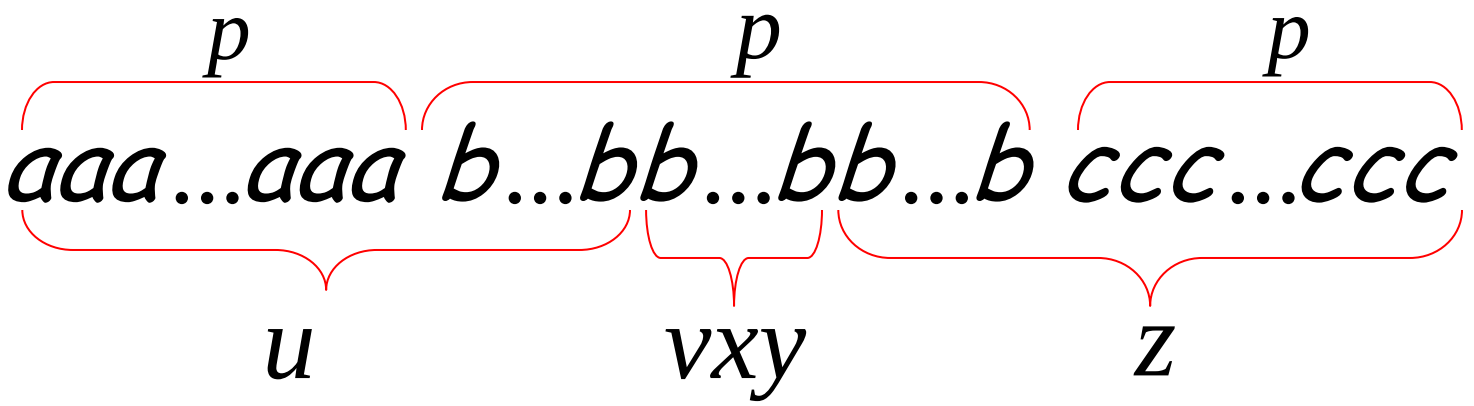
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 2: vxy is in b^p

Similar to case 1



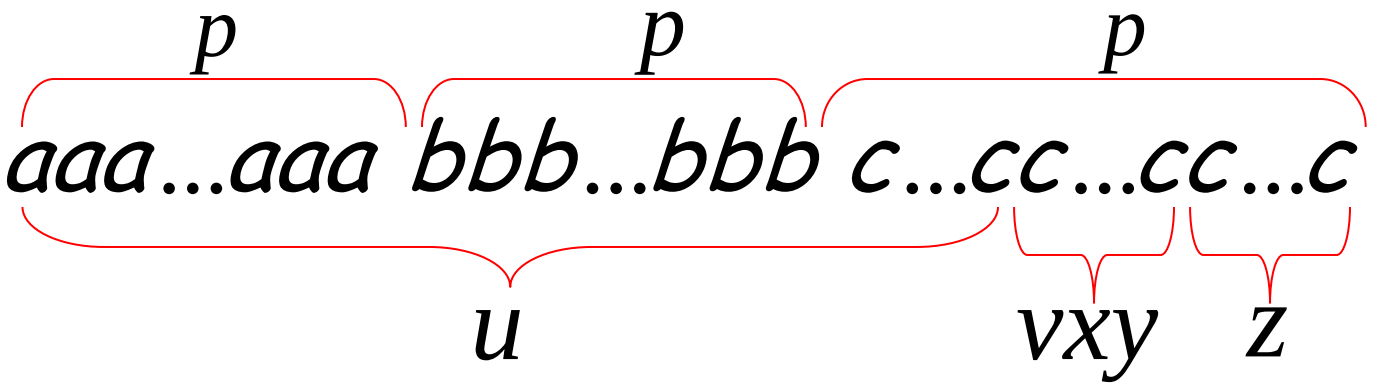
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 3: vxy is in c^p

Similar to case 1

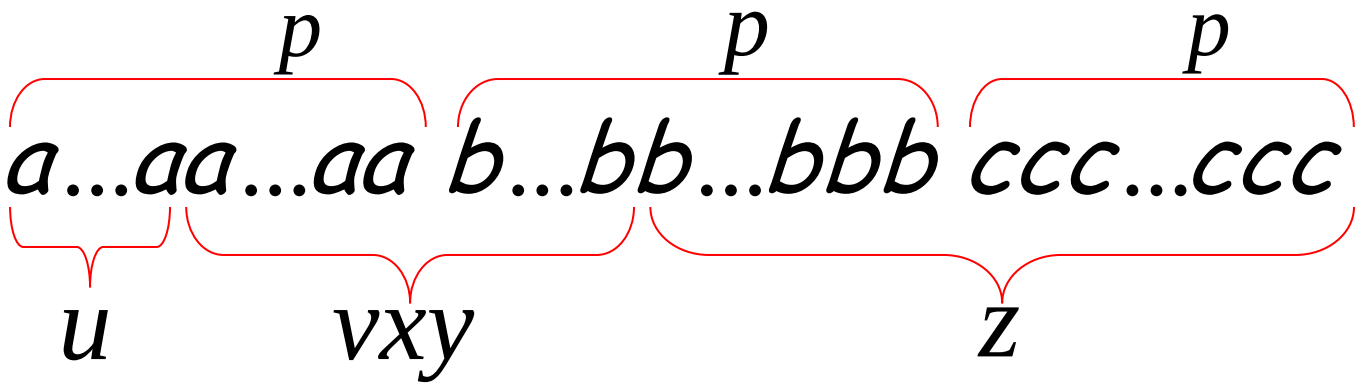


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 4: vxy overlaps a^p and b^p



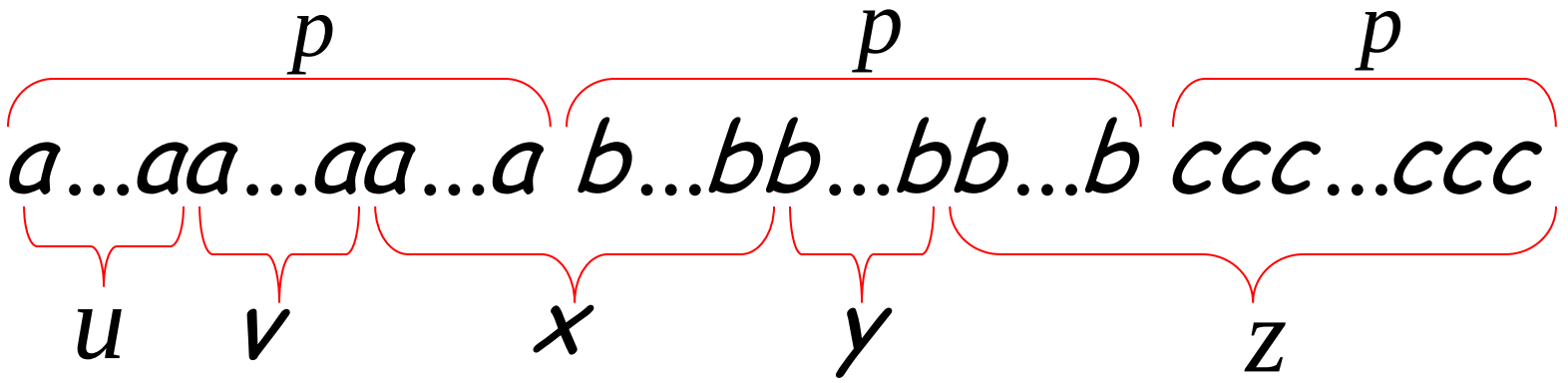
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Sub-case 1: v contains only a

y contains only b

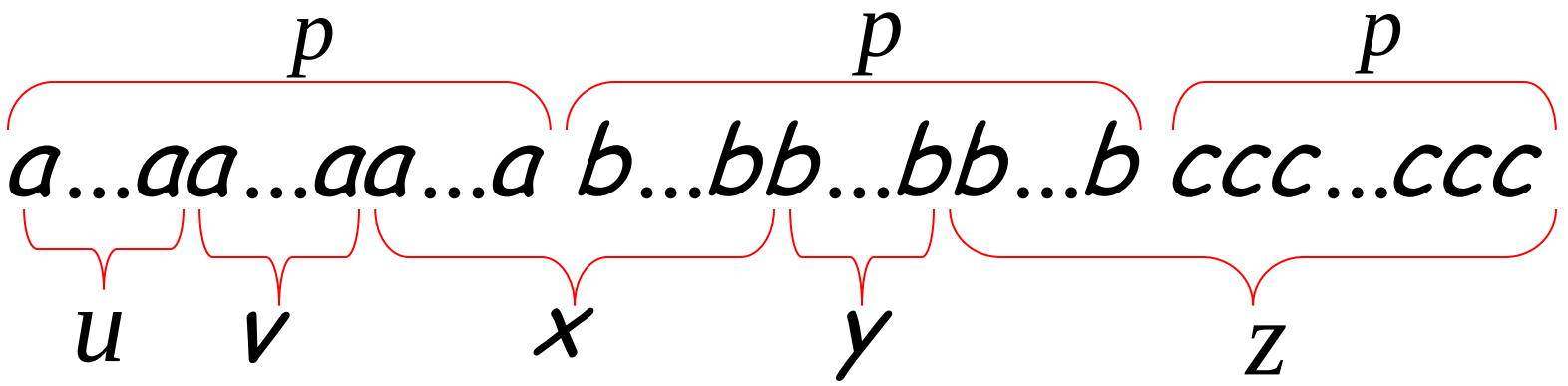


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

$$v = a^{k_1} \qquad y = b^{k_2} \qquad k_1 + k_2 \geq 1$$

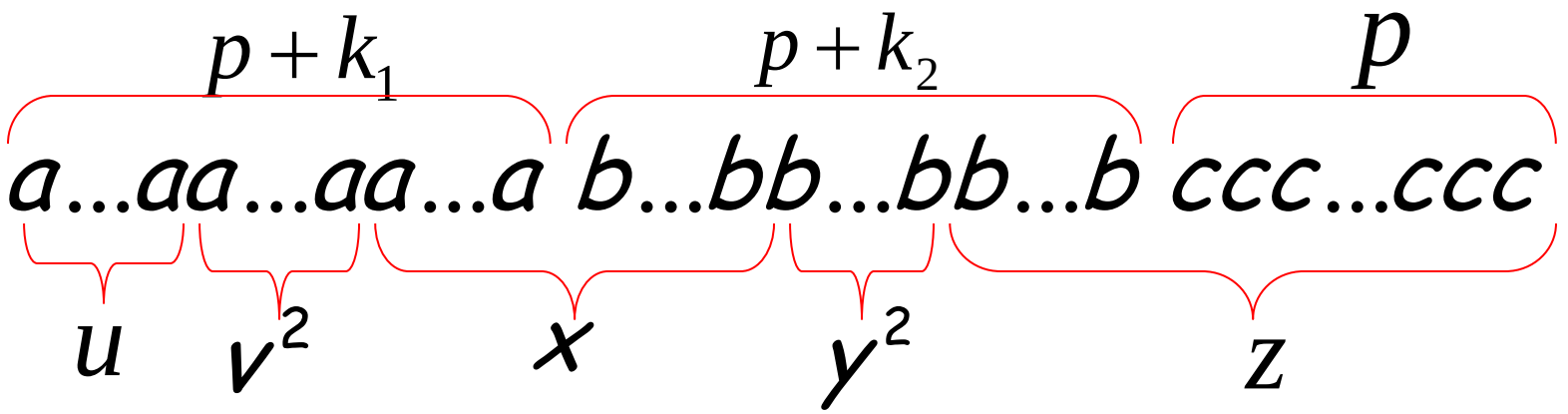


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

$$v = a^{k_1} \qquad y = b^{k_2} \qquad k_1 + k_2 \geq 1$$



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1 + k_2 \geq 1$$

However: $uv^2xy^2z = a^{p+k_1} b^{p+k_2} c^p \notin L$

Contradiction!!!

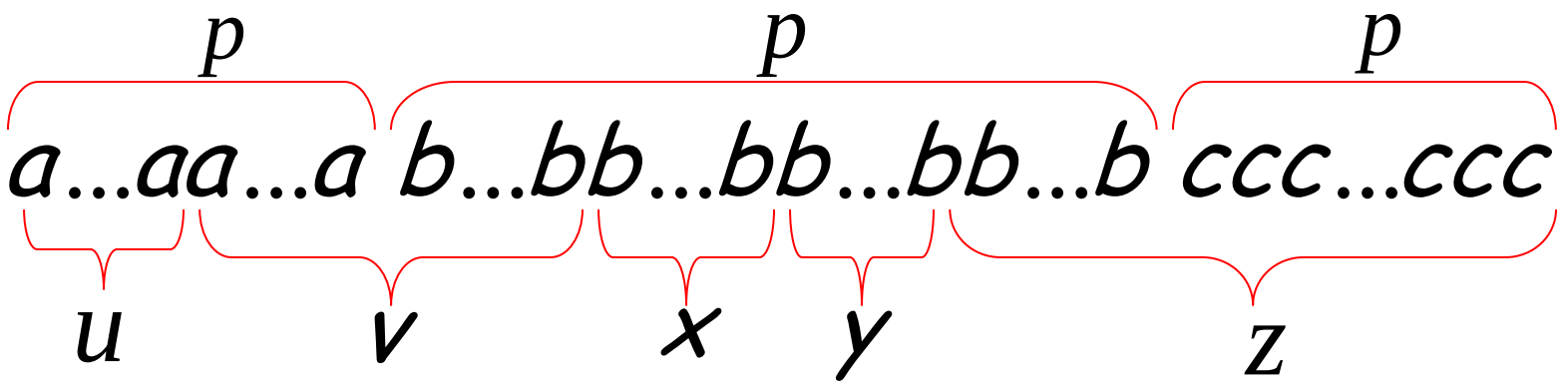
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Sub-case 2: v contains a and b

y contains only b



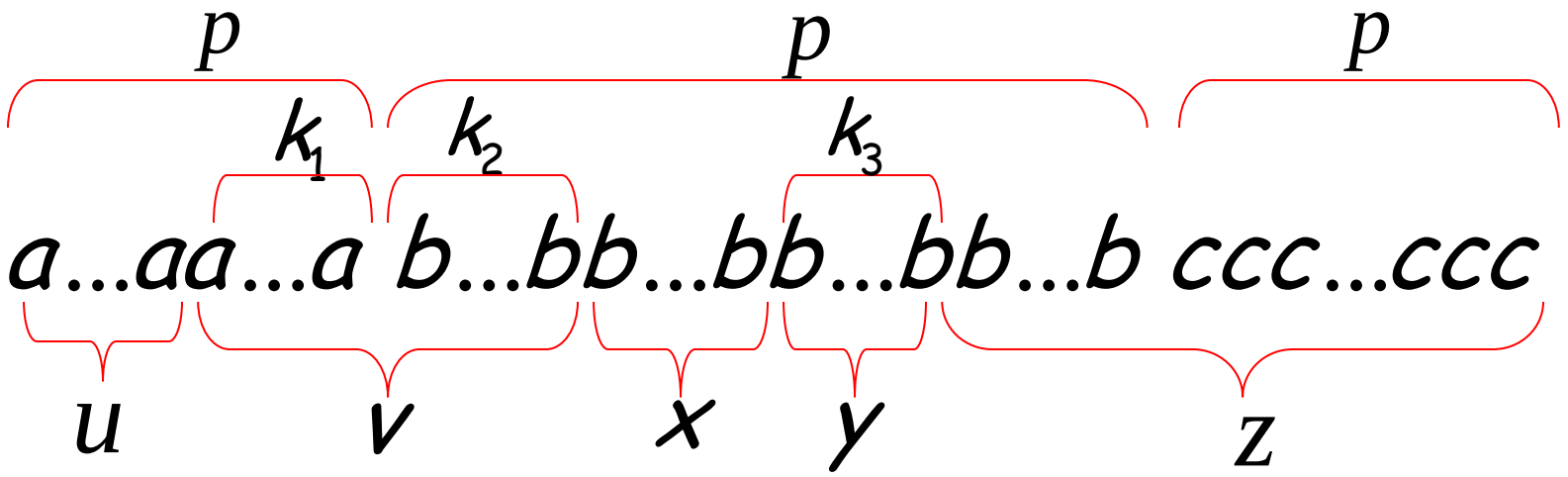
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

By assumption

$$v = a^{k_1} b^{k_2} \qquad y = b^{k_3} \qquad k_1, k_2 \geq 1$$

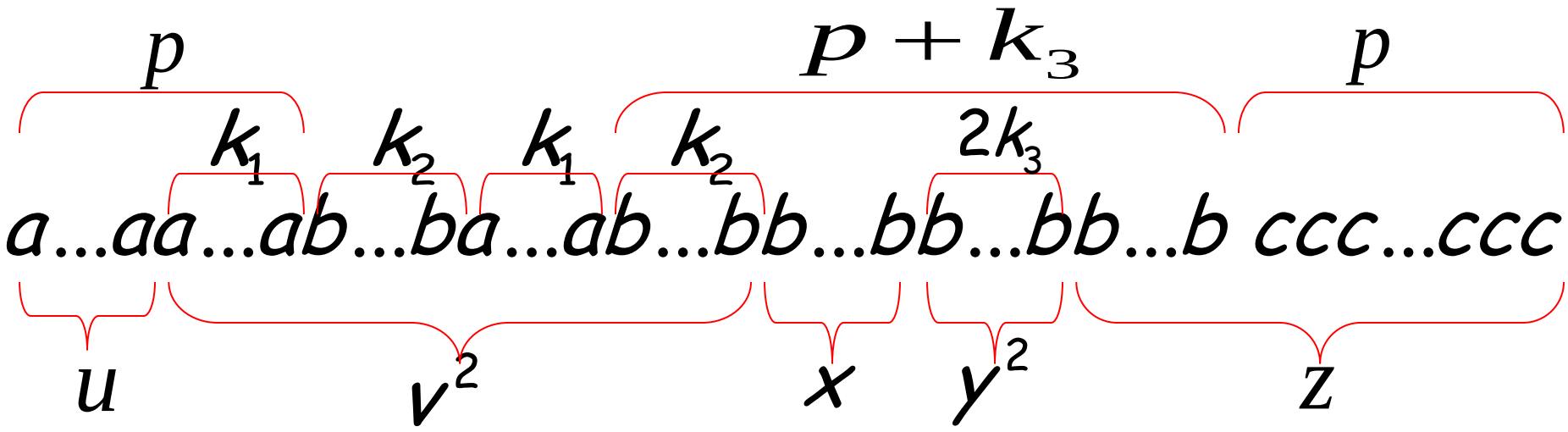


$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

$$v = a^{k_1} b^{k_2} \qquad y = b^{k_3} \qquad k_1, k_2 \geq 1$$



$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

From Pumping Lemma: $uv^2xy^2z \in L$

$$k_1, k_2 \geq 1$$

However: $uv^2xy^2z = a^p b^{k_2} a^{k_1} b^{p+k_3} c^p \notin L$

Contradiction!!!

$$L = \{a^n b^n c^n : n \geq 0\}$$

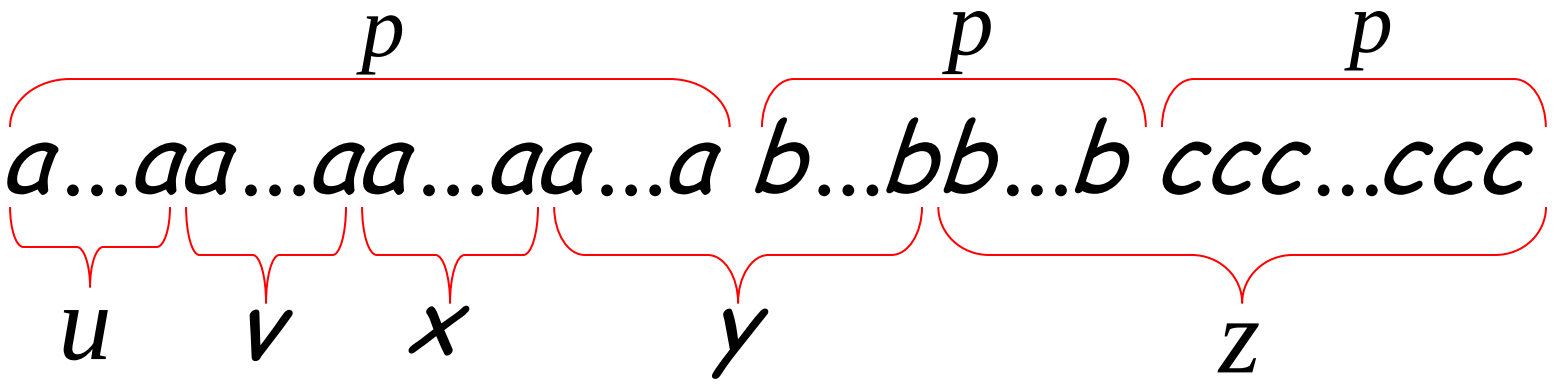
$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Sub-case 3: v contains only a

y contains a and b

Similar to sub-case 2



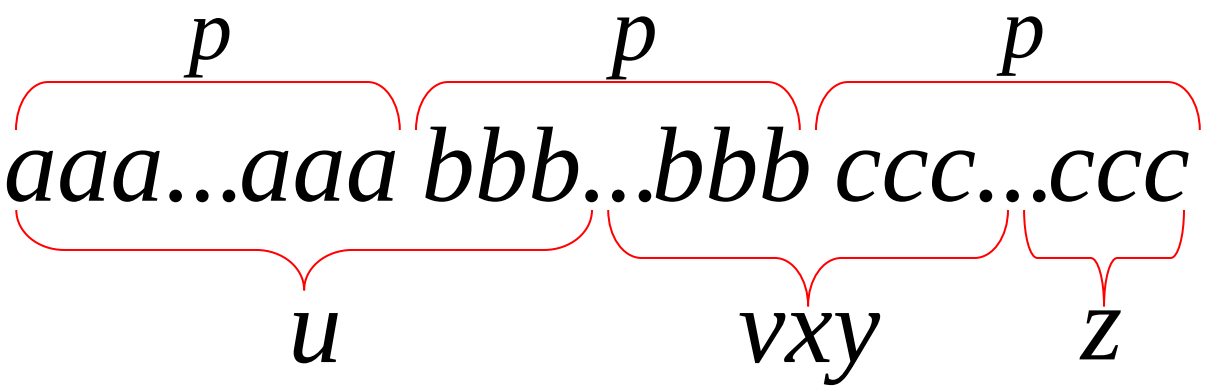
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 5: vxy overlaps b^p and c^p

Similar to case 4



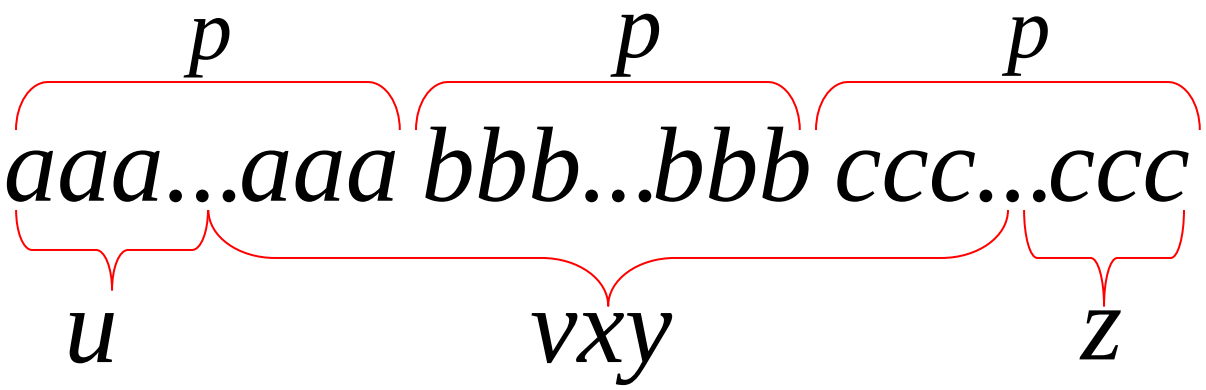
$$L = \{a^n b^n c^n : n \geq 0\}$$

$$w = a^p b^p c^p$$

$$w = uvxyz \qquad |vxy| \leq p \qquad |vy| \geq 1$$

Case 6: vxy overlaps a^p, b^p and c^p

Impossible!



In all cases we obtained a contradiction

Therefore: the original assumption that

$$L = \{a^n b^n c^n : n \geq 0\}$$

is context-free must be wrong

Conclusion: L is not context-free